Reverse Foundations of Physics

Separating Physical Postulates, Mathematical Carriers, and Logical Strength

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Abstract

What part of a physical theory is physics, what part is mathematics, and what part is inherited from the logic and existence principles in which the mathematics is developed? Familiar foundational programmes usually vary one of these coordinates while holding the others fixed. Reverse mathematics calibrates the axioms needed for a theorem. Reverse physics seeks physical postulates sufficient to recover a law. Operational reconstructions recover quantum structure from informational principles. Constructive, computable, choice-free, topos-theoretic, and finite approaches each replace selected pieces of the mathematical architecture. None of those methods alone answers the combined question.

We formulate a programme of *reverse foundations of physics* around the typed derivability judgement

$$L + S + M + \mathrm{Enc}(P) \;\vdash\; O.$$

Here L is the logic, S the set, type, or existence theory, M the mathematical carrier and analytic machinery, P the physical postulates after an explicit encoding, and O one sharply delimited obligation. The programme asks which ingredients are sufficient, necessary, equivalent, independent, or avoidable by reformulation. Its practical research atlas has three coarse axes—six foundational regimes, six carrier families, and sixteen physical obligations—giving 576 navigational coordinates. The cube is not an ontology and its coordinates are not presumed independent.

Five case studies show why the typing matters. An explicit mode labelling can avoid basis-selection uses of Choice without making physics "imply ZF." A Krein fundamental symmetry permits explicit normalized states but does not select a physical state. A represented scalar wave evolution is formalizable over RCA_0, while causal Green support remains a distinct theorem. Exact finite graph causality is not continuum Lorentzian causality. And a bounded finite BV contraction can be checked in PRA without lowering any infinite-dimensional quantum claim. These are sufficiency and separation results, not a universal weakest-foundation theorem. We close with a certificate-oriented research protocol and a set of tractable open problems at the interfaces between logic, analysis, and physics.

1 The missing question

A physical theory is normally presented as if its assumptions came in one list. In practice that list mixes several kinds of commitment. A symmetry principle, a causal condition, and a composition rule are physical postulates. A Hilbert space, a distribution space, a Krein product, or a C^*-algebra is a mathematical carrier. Completeness, compactness, selection of bases, spectral resolution, and

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existence of Green operators are mathematical theorems. Classical logic, set existence, actual infinity, and forms of Choice live one level further down. A calculation can use all four layers without saying where one ends and another begins.

This creates two opposite errors. The first is to treat the conventional mathematical formulation as physical content: quantum mechanics is said to *assume Hilbert space*, even when an operational reconstruction derives a Hilbert representation from other principles. The second is to treat the entire standard mathematical background as unavoidable: a proof uses a basis or compactness theorem, and one concludes that the physical law requires the Axiom of Choice. In both directions the inference skips the representation and the base theory.

Reverse mathematics supplies the right discipline for the mathematical side. Its central question is which set-existence axioms are needed to prove a theorem of ordinary mathematics [13]. Reverse physics applies a parallel strategy to physical laws: begin with accepted laws or theory structure and seek physical starting assumptions sufficient to recover them [5]. Operational reconstructions of quantum theory do something closely related inside a chosen formal arena [9, 6]. Yet the results do not compose automatically. A reconstruction may derive finite-dimensional complex Hilbert structure while relying on classical real analysis. A reverse- mathematical result may calibrate an abstract differential equation while saying nothing about which physical representation should encode its data. Computable analysis may construct an operator under a representation without classifying its proof-theoretic strength. A topos formulation may change the internal logic while leaving the bridge to experimental probabilities open.

The question of this paper is therefore not merely "which physical assumptions imply a theory?" nor merely "which axioms prove its mathematics?" It is:

> For a precisely represented physical obligation, which combination of physical postulates, carrier structure, existence principles, and logic is sufficient; which parts can be recovered from the obligation; and which can be removed by changing representation?

We use *reverse foundations of physics* as a working name for this combined programme. The phrase is descriptive, not a priority claim. The adjacent literatures are extensive, and the present literature survey is scoped rather than exhaustive. Our claim is narrower: the sources reviewed here do not provide a single typed atlas in which physical obligations, carrier choices, and logical or set-theoretic strength are all varied and audited together.

The programme grew from conformal gravity because that theory is a demanding stress test. Gauge symmetry, indefinite pairings, fourth-order evolution, Green operators, local BRST cohomology, state selection, renormalized products, and the quantum master equation all occur, but they occur at different dependency levels. The method is not specific to Weyl gravity. Its natural domain includes classical PDE, quantum mechanics, algebraic quantum theory, lattice systems, operational reconstructions, and any theory in which a change of mathematical foundations may change what "existence" or "state" means.

Contributions. This paper makes four contributions. First, it gives a typed formal object for the combined analysis. Second, it organizes a three-axis research atlas and explains what its coverage counts do and do not mean. Third, it presents five independently checkable case studies in which familiar foundational inferences split into distinct claims. Fourth, it proposes an evidence and certificate protocol designed to turn a suggestive matrix cell into a result that can survive changes of proof, representation, and implementation.

2 A typed derivability problem

2.1 Profiles, obligations, and encodings

The basic object is not a theory name but a derivability profile.

Definition 2.1 (Assumption profile)**.** An assumption profile is a tuple

$$\mathcal{A} = (L, S, M, \mathrm{Enc}(P), R),$$

where L specifies the logic and inference rules, S specifies the existence theory, M specifies the mathematical carrier and theorems made available over $L + S$, P is a list of physical postulates, Enc is a formal encoding of those postulates, and R records the representation of inputs and outputs.

The representation R deserves its own coordinate. A real number may be a Dedekind cut, a rapidly converging Cauchy name, or a point of an internal real object. A field may be a smooth section, a distribution, a Sobolev class, a finite lattice function, or a compatible family of finite data. Those representations can change both the content of an existence claim and the strength of its proof. We display R inside the profile but often absorb it into M and Enc, giving the compact judgement

$$L + S + M + \mathrm{Enc}(P) \;\vdash_R\; O. \tag{1}$$

Definition 2.2 (Obligation)**.** An obligation O is one theorem-level job with a declared result kind and scope: for example, construction of one normalized algebraic state; uniqueness of an energy solution in a represented carrier; existence of retarded and advanced Green maps with a support theorem; classification of local counterterms; or restoration of a local quantum master equation.

The obligation must be smaller than a theory. "Construct quantum field theory" is not an obligation. "Construct a BRST-compatible Hadamard state for the full metric BV complex" is an obligation, albeit a very difficult one. This granularity prevents a finite-mode calculation from being promoted to a continuum field theorem and prevents state existence from being promoted to a state-selection principle.

2.2 Seven relations, not one arrow

An unlabelled implication arrow hides too much. We use the following relation types.

Relation	Meaning
`USED_BY_DISPLAYED_PROOF`	One inspected proof invokes the item. This is neither necessity nor an optimal upper bound.
`SUFFICIENT_OVER_BASE`	A derivation of the obligation is available over the declared base, encoding, and representation.
`NECESSARY_OVER_BASE`	A reversal, separating model, or counterexample proves that the item cannot be omitted over the declared base.
`EQUIVALENT_OVER_BASE`	Both sufficiency and necessity have been proved over the same base and encoding.
`AVOIDED_BY_REFORMULATION`	Another explicit representation reaches the same declared obligation without the item.
`INDEPENDENT_OVER_BASE`	Models or realizations on both sides establish independence over the named base.
`UNKNOWN`	No stronger relation is certified.

These types separate three questions that are often collapsed:

1. Did a proof use a principle?
2. Is the principle sufficient or necessary for the theorem over a named base?
3. Does the physical obligation require the theorem in that representation and generality?

Only the second question is a reverse-mathematical calibration. The third is where reverse mathematics and reverse physics meet.

2.3 Why physical assumptions do not simply imply Choice

The phrase "physical assumption P implies the Axiom of Choice" is usually ill-typed. A physical postulate and a set-theoretic sentence do not even inhabit a shared formal language until an encoding and base theory have been fixed.

Proposition 2.3 (Encoding discipline)**.** *Let P be a physical postulate stated in an experimental or operational vocabulary and let C be a choice principle stated in a foundational language. A claim $P \Rightarrow C$ has no formal derivability content until one supplies a common theory $L + S$, an encoding $\mathrm{Enc}(P)$, and a target translation of C. Any proved implication then belongs to that encoding and base; it is not automatically an implication from an experiment to an unrestricted assertion about all sets.*

This does not make such reversals impossible. A physical principle may be encoded as a totality, selection, compactness, or continuity statement, and that statement may turn out to be equivalent to a familiar subsystem or choice fragment. The proposition says what must be provided before the claim is meaningful.

Physical structure can nevertheless reduce mathematical commitments in a more direct way. If the physics supplies a canonical grading and explicit mode labels, a proof need not choose a basis. If it supplies a preferred state or boundary condition, a selection problem may disappear. The correct relation in such cases is often `AVOIDED_BY_REFORMULATION`, not "physics proves Choice" or "physics disproves Choice."

3 The research atlas

The formal profile has many coordinates. For navigation we project it onto three axes: foundational regime, carrier, and physical obligation. The current projection has $6 \times 6 \times 16 = 576$ coordinates. This number is large enough to expose blind spots and small enough to inspect. It is not a claim that the underlying space of theories is finite, that the three axes are independent, or that every coordinate is coherent.

3.1 Six foundational regimes

Regime	Intended question
Classical standard	What is known in ordinary classical ZFC-style analysis, where background Choice is often not audited?
Weak arithmetic	How much proof-theoretic strength is needed in systems such as PRA, RCA_0, WKL_0, or ACA_0?
ZF with weakened Choice	Which constructions survive in classical set theory when full Choice or Countable Choice is absent or isolated?
Constructive/computable	What witnesses, algorithms, moduli, or represented operators can be constructed without treating classical existence as enough?
Topos/internal	What becomes of the theory when objects and truth are interpreted internally, often with Heyting rather than Boolean logic?
Finite/discrete restriction	What survives on finite carriers, exact finite algebras, lattices, or proposals that restrict actual infinity?

These regimes overlap but are not synonyms. ZF without Choice retains classical logic and actual infinity. Bishop-style constructive mathematics changes the meaning of existence. Computable analysis fixes representations and algorithms but is not by itself a subsystem classification. A topos has an internal logic relative to a selected category. A finite cutoff inside ZFC is not finitism.

3.2 Six carrier families

Carrier	What it is meant to track
Finite exact algebra	Finite matrices, integer or rational complexes, and explicit finite-dimensional algebraic data.
Hilbert/operator	Positive Hilbert spaces, completions, operator domains, spectral data, and one-parameter groups.
Krein/indefinite	Indefinite pairings, fundamental symmetries, positive companion topologies, and their operator theory.
Algebraic C^*-system	Observable algebras, states, GNS representations, nets, and algebra-first dynamics.
Smooth/PDE/ distributional	Manifolds, sections, Sobolev and distribution spaces, differential operators, and Green theory.
Localic/synthetic/ internal	Locales, internal algebra objects, formal manifolds, and synthetic smooth structures.

The rows are deliberately not mutually exclusive descriptions of a complete theory. A Krein space normally carries a positive companion Hilbert topology; an algebraic quantum theory may be represented on a Hilbert space; and a PDE may be encoded in a finite exact approximation. The axis asks which carrier is doing the work in the obligation under inspection.

3.3 Sixteen obligations in four groups

The obligation axis is the most important because it blocks silent promotion. For readability the sixteen values fall into four groups.

Group	Obligations
Kinematics and states	kinematics/observables; state existence; state representation; probability rule; physical state selection.
Dynamics and causality	generator/spectral dynamics; evolution and well-posedness; causal propagation and advanced/retarded Green maps.
Gauge and interaction	gauge/BV/cohomology; interaction construction; counterterm classification; anomaly classification.
Quantum completion and comparison	renormalized products; QME restoration; residual quantum transfer; reconstruction and continuum or empirical limits.

This ordering encodes no claim that every theory must pass through the stages linearly. It does show why "we have a state" does not answer which state is physical, why a unitary group does not supply a retarded propagator, and why a local anomaly classification does not restore a quantum master equation.

3.4 Coverage is metadata, not truth

In the current evidence snapshot, 452 of the 576 coordinates are emitted for inspection. Among those, 93 are marked as literature results, 88 as local results, 160 as having only pieces, 30 as priority gaps, and 81 as reviewed but not mapped. The remaining 124 are synthetic complements not yet emitted as coherent coordinates. Thus 205 positions lack substantive coverage in the current explorer. The evidence interface contains 69 source or certificate records.

Those numbers measure the state of a corpus and classification scheme. They do not measure how much humanity knows, and an unfilled cell is not evidence of impossibility. A literature-result label says that a source establishes a claim with a recorded boundary; it does not say the entire cell is solved. A local-result label says that a repository certificate reaches the declared obligation; it does not make the result representation-independent. A priority gap is an invitation to research, not a no-go theorem.

The counts are reproducible from `FOUNDATIONAL_INTERSECTION_CUBE_V4` and `FOUNDATIONAL_MATRIX_EXPLORER_SITE_V2`. The cube explicitly records that it is not literature complete, does not assess all 576 coordinates, proves no weakest base or reversal for the coded wave result, and establishes no new Lorentzian Weyl claim.

4 How adjacent programmes fit—and fail to collapse

4.1 Reverse mathematics

Reverse mathematics most directly calibrates the $L + S \vdash T$ part of Eq. (1). Its strength is the insistence on a named weak base and, in a full reversal, proofs in both directions. Simpson's analysis of the Cauchy–Peano theorem is an instructive physics-adjacent example: over RCA_0, the theorem is equivalent to WKL_0, while the Ascoli lemma and Osgood's maximum-solution theorem have the strength of ACA_0 [12]. "Differential equations require analysis" is thereby replaced with several exact statements whose strength depends on the theorem.

The limitation for our purpose is equally clear. Reverse mathematics begins after a statement and its representation have been selected. It does not by itself decide whether the physical problem requires Cauchy–Peano in that generality, whether a canonical physical flow avoids the compactness step, or whether a different carrier expresses the same observable content.

4.2 Reverse physics and operational reconstruction

Reverse physics begins from laws and seeks physical assumptions sufficient to rederive them [5]. This is the closest methodological relative of the present programme. Our additional demand is to audit the mathematical and logical substrate of every displayed derivation.

Quantum reconstructions show why this matters. In Hardy's five-axiom derivation, continuous reversible transformations are load-bearing: deleting continuity yields classical probability theory within the stated framework [9]. Chiribella, D'Ariano, and Perinotti define a broad operational-probabilistic class using five axioms and use purification to single out quantum theory [6]. These works turn some textbook mathematical postulates into derived representation theorems. They do not, without a separate audit, classify the proof-theoretic cost of the continuum, compactness, convex separation, or infinite-dimensional limit.

Accordingly, "Hilbert space is derived" and "Hilbert-space quantum field theory is constructible over a weak or choice-free base" are very different claims. The former can be a finite-dimensional operational theorem; the latter includes completion, unbounded operators, distribution theory, locality, states, and renormalization.

4.3 Choice-free functional analysis

It is common to summarize the situation as "functional analysis needs Choice." Recent results make that too coarse. Blackadar, Farah, and Karagila study Hilbert spaces and operators in ZF without Countable Choice, showing both robust canonical constructions and unfamiliar pathologies [4]. Blackadar and Farah develop substantial separable C^*-algebra theory in ZF, including representation theorems, spectral mapping for polynomials, and continuous functional calculus for commuting normal elements, while also exhibiting failures outside protected settings [3].

The lesson is not that Choice is irrelevant. It is that the cost is attached to a theorem, a class of objects, and a presentation. Canonically presented separable structures may avoid selections that arise for arbitrary families. The right unit of analysis is therefore not "Hilbert space" or "C^*-algebra" but the exact construction used in a physical obligation.

4.4 Constructive, computable, and proof-mined analysis

Constructive analysis requires existence statements to carry witness content [2]. Computable analysis makes that demand relative to explicit representations of spaces and operators [14]. Weihrauch and Zhong, for example, give an algorithm for distributional fundamental solutions of constant-coefficient differential operators under canonical representations [15]. That result is highly relevant to the carrier and computability axes, but a fundamental solution is not automatically a retarded or advanced solution and the theorem does not by itself supply Lorentzian causal support.

Proof mining asks a different question again: what quantitative or computational information can be extracted from an apparently nonconstructive proof? Pischke's metatheorem for nonlinear semigroups generated by accretive operators illustrates its reach into PDE and evolution, including extracted rates [11]. Proof mining should not be relabelled as a Big-Five reverse-mathematical classification, nor should a computable realizer be silently treated as a constructive proof in every foundational system. The three methodologies can inform one another only after their input and output types are recorded.

4.5 Topos and localic reformulations

Constructive Gelfand duality replaces point-set spectra with point-free spectra in a form valid constructively [7]. Heunen, Landsman, and Spitters place a noncommutative algebra in a topos of its commutative contexts, where the internal algebra is commutative, its spectrum is a locale, and the internal propositional logic is intuitionistic [10]. This is a genuine change of mathematical architecture, not merely deletion of Choice from a classical proof.

It also illustrates a recurring boundary. Internal state objects, locale valuations, and internal dynamics do not automatically supply a physical state-selection rule, a Lorentzian Green operator, an interacting BV complex, or empirical equivalence with the external formulation. Object logic and metalogic must be recorded separately. A non-Boolean algebra of quantum propositions is not, just by itself, an intuitionistic metatheory.

4.6 Finite mathematics and finite physics

Finite structures offer exact computation and sometimes eliminate analytic existence questions. But "finite" has several types. Gibbons, Hoffman, and Wootters use a finite field to label a discrete phase space when the quantum state-space dimension is a prime power [8]. The carrier of state vectors remains an N-dimensional complex Hilbert space. A finite field phase space is therefore not the same thing as a finite-field amplitude theory, a finite-mode truncation, or a rejection of actual infinity.

The distinctions become decisive when a finite regulator is used to motivate a continuum claim. Exact solvability at each finite size supplies a family of finite theorems. A continuum limit requires its own representation, uniform estimates, convergence theorem, and physical comparison. Finitism is a foundational proposal; a lattice cutoff inside ordinary mathematics is an approximation scheme.

5 Five certified separations

The following examples are not offered as a complete theory. They are small enough to verify independently and together show what the combined programme changes in practice.

5.1 Krein structure changes the pairing, not all Hilbert mathematics

A Krein space carries a nondegenerate indefinite product together with a fundamental decomposition or fundamental symmetry. If $J = J^* = J^{-1}$, the companion product

$$(x, y)_J = [x, Jy]$$

is positive and supplies Hilbert topology. Replacing a positive physical pairing by a Krein pairing therefore does not remove completion, topology, or operator-domain questions. It redistributes them.

In the reduced conformal-gravity mode carrier audited in `FOUNDATIONAL_KREIN_EXPLICIT_J_ZF_V1`, energy, family, chirality, and a finite multiplicity coordinate name every mode. The fundamental symmetry is coordinate multiplication by the sign of the family. At every fixed cutoff, the list and the involution are primitive-recursive finite data; PRA suffices to verify self-adjointness and $J^2 = 1$. At the complete one-particle level, the carrier is an explicitly indexed $\ell^2(I)$, and the occupation number Fock carrier is $\ell^2(\mathrm{Occ}(I))$. The certificate uses ZF results for those canonical completions and no Countable Choice.

The foundational work is done by the representation. Named modes replace an unspecified basis search and a choice of bases across arbitrary families. The certified relation is `AVOIDED_BY_REFORMULATION` for those particular selection steps and `SUFFICIENT_OVER_BASE` for ZF. It is not a proof that arbitrary Krein spaces have choice-free fundamental decompositions or that ZF is the weakest possible base.

This case also corrects the slogan "Krein throws out Hilbert space." The distinguished physical pairing becomes indefinite, but a positive companion Hilbert topology remains part of the displayed structure. What changes are the adjoint relations, positivity problem, null directions, and physical interpretation; what survives must still be audited theorem by theorem.

5.2 State existence is not state selection

Once the coordinate Krein carrier is available, constructing *a* state is easy. A named J-eigenvector e_i defines an explicit normalized vector state on the companion Hilbert algebra,

$$\omega_i(A) = (e_i, Ae_i)_J,$$

and the rank-one density operator $|e_i\rangle\langle e_i|$ is explicit. No choice principle is required for this witness. The same holds for named occupation vectors in the Fock carrier. These constructions are certified by `FOUNDATIONAL_KREIN_STATE_SELECTION_ZF_V1` with dependency tags `LOCAL-ALGEBRAIC` and `REDUCED-MODE`.

But J does not choose a unique coordinate. Indeed, the certificate proves a scoped no-selection result: no density operator is invariant under the full group of sign-preserving finite-coordinate permutations on the infinite positive and negative sectors. The reason is elementary. Invariance makes all diagonal weights within an infinite sector equal; trace-class normalization then forces each equal weight to vanish, contradicting unit trace. This excludes normal density states with that full invariance. It does not exclude singular states, and it does not prove that the permutation group is a required physical symmetry.

Thus three cells that are often merged must be separate:

$$\text{state existence} \quad \not\Rightarrow \quad \text{canonical state selection} \quad \not\Rightarrow \quad \text{Born interpretation.}$$

Extra physical information—dynamics, boundary conditions, thermality, Hadamard regularity, detector preparation, or an incoming condition—must do the selecting. The algebra and its set-theoretic foundation cannot supply that information merely by existing.

A related audit, `FOUNDATIONAL_BT_SEPARABLE_STATE_CHAIN_ZF_V1`, reaches the same boundary from an algebra-first direction. An explicitly presented separable compact- operator unitization, concrete states, and a corner GNS representation can be constructed in ZF. A semifinite trace is not a normalized state, and conditioning on a detector projection imports that projection as additional physical data. Algebra, state existence, representation, and physical selection are four obligations.

5.3 Weak wave evolution is not causal Green theory

The certificate `FOUNDATIONAL_CODED_POLYGONAL_WAVE_RCA0_V1` treats a mean-zero scalar wave on a coded circle. Dense states are pairs of rational periodic step functions whose polygonal primitives encode left- and right-moving profiles. Completed states and real times are supplied as fast Cauchy names with fixed rates. Rational translation is exact and preserves the energy metric. A finite-code time modulus and a primitive-recursive diagonal construction extend the translation group to named real times.

For this representation, RCA_0 suffices for existence, uniqueness, the group law, and energy conservation of the continuous isometric extension. Finite rational-code identities are already checkable in PRA. This is a genuine weak-base upper bound, and it demonstrates how much can be obtained when convergence rates are data rather than objects to be extracted from bare convergence.

The boundary is as important as the theorem. The result proves neither that RCA_0 is necessary nor that the upper bound is representation-invariant. It constructs no spacetime distribution, no advanced or retarded Green map, and no finite-propagation theorem. A unitary or isometric evolution group solves a Cauchy obligation; causal support is a separate spacetime theorem.

The literature makes the missing bridge visible. Normally hyperbolic operators on globally hyperbolic spacetimes have advanced and retarded Green operators with support control in classical analysis [1]. Under represented hypotheses, parts of PDE and fundamental-solution theory are computable [15]. The current audit has not found a direct reverse-mathematical reversal, a Bishop-style global Green theorem, or a choice-free ZF audit for the full variable-coefficient Lorentzian construction. "Classically known," "computably represented," and "calibrated over a weak base" are three different statuses.

5.4 Exact finite causality is not continuum causality

On a finite nearest-neighbour graph, a discrete wave recurrence can be solved by exact rational arithmetic. The retarded response at time step n is a polynomial in the finite propagation operator of degree at most n. Consequently its matrix entry vanishes when graph distance exceeds n. The advanced kernel follows by time reversal. The construction and graph-step support theorem are certified in `FOUNDATIONAL_FINITE_GRAPH_WAVE_CAUSALITY_V1`.

This is a clean causal theorem, but its type is `LOCAL-ALGEBRAIC/REDUCED-MODE`, not `LORENTZIAN-CAUSAL`. It does not provide a continuum Green operator, a light-cone support theorem, stability under refinement, a regulator- independent limit, or a metric BV propagator. Each of those requires new mathematical data.

The example gives a useful rule for finite programmes:

$$\begin{gathered}\text{exact finite identity} + \text{uniform estimate}\\ + \text{represented convergence} + \text{limit identification}\\ \implies \text{candidate continuum theorem.}\end{gathered}$$

Only the first term is supplied by exact finite causality. This does not make the finite theorem unimportant. It tells us exactly which part of continuum causality is algebraic and which part is analytic.

5.5 A finite BV contraction does not freeze the quantum theory

The energy-two free BV block provides another finite exact case. The certificate `FOUNDATIONAL_FREE_BV_ENERGY2_PRA_SDR_V1` verifies integer matrices for a strong deformation retract and checks the required identities in PRA. Because the contraction and projection are supplied explicitly, no Hahn–Banach extension or abstract complement selection is needed for this fixture.

The result is intentionally small. It does not verify all energies, the field-derived infinite complex, the Fock lift, a Green operator, a state, or a renormalization construction. It does not pass the classical import freeze gate for the quantum programme. The dependency tag is exactly `LOCAL-ALGEBRAIC`. A bounded exact certificate can remove foundational ambiguity from one algebraic step while leaving every analytic and causal obligation open.

This is the intended interaction with the Weyl BV–BFV programme. The classical complex is authoritative, but its import into quantum work must be checked object by object. Local counterterm and anomaly classification must precede coefficient calculations; restoration of the local quantum master equation must precede residual transfer. No reduced-mode or Euclidean result may be used as evidence for a Lorentzian causal claim. The reverse- foundational atlas makes these dependency cuts visible before a calculation is promoted.

6 What the case studies teach

The examples support several general lessons. They are methodological conclusions from the displayed certificates, not universal theorems about all physical theories.

6.1 The first cost of infinity is often completion, not Choice

In explicit separable carriers, countable infinity can enter through a named index set and canonical completion without any selection from an arbitrary family. The finite-to-infinite boundary still adds real mathematical content: summability, completion, operator domains, and limits. It simply need not add Countable Choice in the same step. “Infinite” and “choice-dependent” are not synonyms.

6.2 Physical input often acts as data, not as a foundational axiom

A mode grading, detector projection, boundary condition, or convergence modulus can remove an existential search. The physical postulate does not thereby become a set-theoretic axiom. It supplies a witness inside the chosen foundation. This is one of the most promising ways physics can lower the mathematical strength of a formulation: replace arbitrary existence with canonical physical data.

6.3 Representation sensitivity is scientific content

The RCA_0 wave theorem works because fast Cauchy rates and rational polygonal dense data are part of the representation. That is not a defect to hide. It tells us precisely which information makes the evolution effective and weakly formalizable. A later theorem may show that another natural representation is equivalent over the same base, or it may expose a genuine difference. Until then, representation invariance is an open obligation.

6.4 Positive structure migrates rather than disappears

Krein geometry removes positive definiteness from the distinguished pairing but restores a positive companion topology through a fundamental symmetry. Algebra-first formulations remove state vectors from the primitive vocabulary but recover Hilbert representations through GNS. Operational reconstructions remove Hilbert space from the primitive postulates but recover it from continuity, composition, and informational principles. In each case the assumption load moves. An honest analysis follows it.

6.5 The difficult cells are interfaces

Finite algebra, explicit coordinate carriers, and named states are relatively tractable. The sparse region begins where several interfaces must be crossed at once: weak or constructive analysis plus variable-coefficient Lorentzian PDE; internal topos structure plus external experimental probability;

gauge cohomology plus microlocal renormalization; or an operational reconstruction plus infinite-dimensional continuum limits. These gaps are not random. They mark where entire mathematical toolchains must be retyped.

7 A research protocol

The matrix becomes scientifically useful only if filling a cell means more than attaching a citation. We propose the following protocol.

7.1 Declare the target before auditing the proof

Every project begins with one obligation, one carrier, one representation, and one dependency boundary. "Wave existence" should say whether the solution is a Cauchy name, a Sobolev function, a distribution, or a smooth field. "State" should say algebraic state, normal density state, KMS state, Hadamard state, or physical selector. "Causality" should say graph-step support, finite propagation, or advanced/retarded Green support.

7.2 Build a proof ledger

The displayed proof is factored into individual dependencies: finite algebra, completion, compactness, extension, subsequence selection, spectral calculus, operator-domain closure, Green existence, support, duality, and so on. Each dependency gets one of the seven relation types. A proof use is never promoted to necessity.

7.3 Search for avoidance before proving a lower bound

Before attempting a reversal, ask whether the physical representation supplies a witness or canonical enumeration. An explicit diagonal operator may avoid a general spectral theorem. A named mode basis may avoid a basis-selection principle. Chiral coordinates may turn a PDE evolution into translation. This often produces a useful upper bound quickly and exposes the precise residue for a later lower bound.

7.4 Separate producer and checker

Exact finite claims should carry machine-readable witnesses and an independent checker that does not merely rerun the producer. Analytic claims should pin the theorem statement, hypotheses, representation, and source. Every result records what would falsify it, what was not checked, and which stronger claim it does not establish. Exact and numerical evidence remain distinct types.

7.5 Promote only through typed gates

For quantum Weyl work, the dependency tags

`LOCAL-ALGEBRAIC`, `EUCLIDEAN-SPECTRAL`, `REDUCED-MODE`, `LORENTZIAN-CAUSAL`

are noninterchangeable. Likewise the lifecycle states

`CLASSIFIED`, `COEFFICIENT_COMPUTED`, `QME_RESTORED`, `RESIDUAL_TRANSFERRED`,
`LORENTZIAN_CERTIFIED`

must not be collapsed. The atlas should show missing typed edges, not draw an unlabelled arrow around them.

8 Near-term research programme

The most valuable next problems are not simply the emptiest cells. They are cells for which one can isolate a small theorem that connects two mature literatures.

8.1 Weak localized wave identities

The represented Cauchy evolution over RCA_0 is established, while spacetime localization is not. A tractable next target is a weak integrated wave identity for compactly supported rational polygonal tests, followed by a support statement for a carefully coded (1+1)-dimensional model. This would test exactly where the move from energy evolution to spacetime distributions adds strength, without beginning with the full metric BV propagator.

8.2 A ZF audit of one Green construction

Choose one normally hyperbolic scalar operator on an explicitly presented globally hyperbolic spacetime. Factor the classical proof into Sobolev completion, energy estimate, local solution, exhaustion, uniqueness, duality, and support. For each step determine whether the standard proof merely uses Choice, whether a separable canonical formulation avoids it, or whether a choice fragment is genuinely needed. The first deliverable should be a dependency ledger, not a claim about all hyperbolic PDE.

8.3 Proof strength of an operational reconstruction

Select one finite-dimensional reconstruction, preferably the continuity branch in Hardy or the purification branch of Chiribella–D'Ariano–Perinotti. Formalize the exact real-algebraic and compactness principles used. Then separate the finite theorem from any proposed countable or field-theoretic extension. This is a promising place to obtain an actual equivalence or avoidance theorem because the physical postulates are already explicit.

8.4 Internal BV and the external probability bridge

The topos literature supplies internal commutative algebra, locale spectra, state representations, and versions of dynamics. A useful Weyl target is smaller than "construct QFT in a topos": internalize a finite local BRST complex and verify nilpotency and a finite contraction, then state precisely what additional structure would be required for integration modulo boundary terms, anomaly classification, and external probabilities. Even an obstruction ledger would convert a vague gap into typed missing morphisms.

8.5 Physical selectors on explicit state spaces

The state-existence problem is already easy in several explicit ZF carriers. The next scientific question is which physical assumptions select a state. Candidate selectors include a Hamiltonian ground condition, KMS condition, incoming detector preparation, residual symmetry, or a local Hadamard condition. Each should be tested independently for existence, uniqueness, foundational strength, and compatibility with gauge structure. A no-selection theorem is valuable only when its symmetry and state class are physically justified.

9 Limits of the present paper

This paper introduces a programme and reports bounded examples. It does not establish a global metatheorem relating physical postulates to set-theoretic strength. It does not prove that the three atlas axes form a literal Cartesian product of coherent theories. It does not prove that the six regimes, six carriers, or sixteen obligations are canonical or exhaustive.

The literature survey is a structured seed corpus, not a systematic review of all mathematical physics. "No reviewed source" means no source in the declared corpus, not that no result exists. The 576-cell count and coverage statistics describe a versioned research instrument. They are not novelty counts, evidence weights, or probabilities that a theorem is true.

The local foundational results are upper bounds unless a reversal is stated. The RCA_0 coded-wave theorem does not show RCA_0 is weakest. ZF sufficiency for the explicit Krein and separable C^* carriers is not a necessity theorem. Exact finite certificates do not settle infinite completions. Computable or constructive results are not automatically reverse-mathematical classifications.

Finally, none of the case studies constructs a complete Lorentzian off-shell BV propagator, a BRST-compatible Hadamard state for the full metric complex, renormalized Lorentzian time-ordered products, a causal perturbative AQFT, or a Lorentzian quantum-master-equation theorem. The certificates carrying `REDUCED-MODE` or `LOCAL-ALGEBRAIC` tags provide no evidence for those `LORENTZIAN-CAUSAL` obligations.

10 Reproducibility

The paper has a machine-readable claim map generated from current certificate inputs. It pins the source paths and hashes, extracts the atlas dimensions and status counts, checks the exact dependency tags of every local case study, and records the paper's nonclaims. The reproduction commands are:

```
python3 paper/generate_21_reverse_foundations_appendices.py --check
python3 paper/generate_21_reverse_foundations_claim_map.py --check
python3 paper/verify_21_reverse_foundations_claim_map.py
cd paper
pdflatex -interaction=nonstopmode -halt-on-error \
  21-reverse-foundations-of-physics.tex
```

The principal machine-readable authorities are `FOUNDATIONAL_INTERSECTION_CUBE_V4`, `FOUNDATIONAL_MATRIX_EXPLORER_SITE_V2`, `FOUNDATIONAL_KREIN_EXPLICIT_J_ZF_V1`, `FOUNDATIONAL_KREIN_STATE_SELECTION_ZF_V1`, `FOUNDATIONAL_BT_SEPARABLE_STATE_CHAIN_ZF_V1`, `FOUNDATIONAL_CODED_POLYGONAL_WAVE_RCA0_V1`, `FOUNDATIONAL_FINITE_GRAPH_WAVE_CAUSALITY_V1`, and `FOUNDATIONAL_FREE_BV_ENERGY2_PRA_SDR_V1`. The static atlas appendices are generated from `foundations/site/data.json`.

11 Conclusion

The mathematical assumptions of a physical theory are neither inert notation nor a single hidden axiom. They are a structured dependency layer. Changing the carrier can move positivity, completion, and representation obligations. Changing the foundation can alter what counts as existence. Physical postulates can supply canonical data that avoid otherwise arbitrary choices. And the same physical slogan can have different strength under different encodings.

The practical response is to replace undifferentiated questions—"does physics need Choice?", "can Krein replace Hilbert space?", "can finite mathematics recover the continuum?"—with typed judgements of the form

$$L + S + M + \mathrm{Enc}(P) \ \vdash_R \ O,$$

where every symbol is explicit and every arrow has a relation type. The first case studies already change the picture. Explicit infinity need not introduce Choice at its first step. State existence does not select a physical state. Evolution does not establish causal support. Exact finite causality does not establish a continuum light cone. And a finite algebraic BV certificate does not promote an analytic quantum theory.

The resulting atlas is useful precisely because it contains holes. A blank cell is not an absence theorem; it is a request for a well-typed obligation, an encoding, a proof ledger, and an independently checkable result. Filling those cells one theorem at a time offers a realistic route toward understanding not only which physics follows from which physical assumptions, but which mathematics the physics truly needs.

A Static companion to the evidence atlas

The interactive atlas exposes five views of the same normalized evidence: matrix, dimensions guide, typed implications, strength ladder, and evidence catalogue. This appendix preserves their key research content in a citable static form. It is a snapshot, not a replacement for cell inspection, filtering, neighboring-cell comparison, or the complete downloadable dataset.

Website view	Static counterpart
Matrix	Overall counts, a 6×6 three-way coverage roll-up, and status counts for every obligation.
Dimensions guide	All 28 axis options with their non-specialist descriptions.
Implications	The complete typed relation ledger: ten directed edges with their exact assertion and evidence.
Strength ladder	All six cylinder-wave gates, including what each adds, establishes, and leaves open.
Evidence	The complete literature register, local-certificate register, and usage crosswalk for all 69 records.

A.1 Dimensions guide

Every coordinate combines one foundational regime, one carrier, and one theorem-level obligation. The tables below reproduce the website's plain-language guide rather than assuming the reader already knows the programme vocabulary.

A.1.1 Under which rules may the mathematics reason and assert that objects exist?

Option	Plain-language purpose
Classical standard	Mainstream mathematics: classical logic, completed infinite structures, and ordinary analysis, with Choice available unless a proof explicitly avoids it.
Weak formal base	Use a deliberately small formal system and ask exactly how much arithmetic or set existence the proof needs.

Option	Plain-language purpose
ZF with weakened Choice	Keep classical set theory but remove or isolate principles that choose objects from infinitely many sets at once.
Constructive/computable	An existence claim must provide a witness, construction, or algorithm—not only show that nonexistence would be contradictory.
Topos/internal	Do the mathematics inside an alternative logical universe, where truth may be local and classical either/or reasoning may fail.
Finite/discrete restriction	Replace an infinite or continuous system by finite exact data or finitely many modes. This is not automatically the same as rejecting infinity as a foundation.

A.1.2 What mathematical object carries states, observables, fields, and evolution?

Option	Plain-language purpose
Finite exact algebra	Finite matrices, rational arrays, or other finite algebraic data that can be checked exactly.
Hilbert/operator	The positive-norm vector spaces and operators used in standard quantum mechanics and spectral theory.
Krein/indefinite	A vector space whose inner product can be positive, negative, or zero, as often occurs before unphysical gauge directions are removed.
Algebraic C*-system	Start from an algebra of observable quantities; a state is a rule assigning expectation values rather than primarily a wavefunction.
Smooth/PDE/distributional	Continuum fields on space or spacetime, including derivatives, PDEs, Sobolev spaces, generalized functions, and Green operators.
Localic/synthetic/internal	Describe spaces through regions, logical relations, or internal geometry instead of beginning with a set of individual points.

A.1.3 Which precise physical or theorem-level job is established?

Option	Plain-language purpose
Kinematics/observables	Say what the possible configurations and measurable quantities are before specifying how they evolve.
State existence	Show that at least one mathematically valid state actually exists.
State representation	Explain how an abstract state is encoded—for example by a vector, density matrix, measure, valuation, or GNS construction.
Probability rule	Turn states and events into normalized probabilities, such as a Born-type prediction rule.
Physical state selection	Explain why a particular vacuum, thermal, Hadamard, or other state should count as physically distinguished.
Generator/spectral dynamics	Construct what generates time evolution and, where relevant, identify its allowed frequencies or energy spectrum.
Evolution/well-posedness	Show that admissible initial data produce a solution that exists, is unique, and changes stably or computably with the data.
Causal propagation/Green	Show that disturbances propagate within the permitted causal region and construct retarded or advanced response maps.
Gauge/BV/cohomology	Handle redundant gauge descriptions consistently and identify the quantities or states that remain physically meaningful.
Interaction construction	Build a genuine coupling or nonlinear theory rather than only a collection of free, noninteracting fields.

Option	Plain-language purpose
Counterterm classification	List every local correction that quantum calculations are allowed to require before attempting to calculate its coefficient.
Anomaly classification	List the possible ways a classical symmetry or consistency condition could fail after quantization.
Renormalized products	Define products and correlation functions that would otherwise be singular when fields meet at the same spacetime point.
QME restoration	Repair the quantum master equation, the BV consistency condition that encodes quantum gauge symmetry.
Residual quantum transfer	After quantum consistency is restored, transfer the correction to the smaller complex that represents the surviving physical content.
Reconstruction/limits	Connect the formulation back to operational predictions, a continuum or standard theory, or a demonstrated notion of empirical equivalence.

A.2 Matrix overview

The full atlas has 576 coordinates. The website exposes sixteen 6×6 slices, one for each obligation. For print, Table 4 aggregates those slices without implying that the statuses are truth values. Each entry is *direct / seeded / unmapped*: direct combines local and reviewed literature results; seeded combines pieces-only and priority-gap cells; unmapped means that no coverage classification is made.

Regime ↓ / carrier →	Finite exact	Hilbert	Krein	Algebraic	PDE	Localic
Classical standard	8/7/1	7/3/6	6/4/6	12/4/0	9/6/1	5/4/7
Weak formal base	9/7/0	3/2/11	3/1/12	0/3/13	0/10/6	0/0/16
ZF with weakened Choice	7/8/1	5/5/6	5/4/7	10/6/0	2/8/6	2/2/12
Constructive/computable	7/8/1	5/5/6	1/3/12	5/3/8	1/9/6	5/4/7
Topos/internal	7/7/2	2/4/10	4/3/9	5/4/7	2/7/7	3/13/0
Finite/discrete restriction	10/6/0	7/3/6	7/8/1	7/8/1	5/7/4	5/4/7

Table 4: Aggregate coverage matrix. Each cell reports direct results / open cells with a starting point / not mapped, across all sixteen obligations.

Table 5: Coverage status by physical obligation. Every row sums to 36.

Obligation	Local	Literature	Pieces	Priority gap	Not mapped	Total
Kinematics/observables	8	17	1	0	10	36
State existence	11	10	6	0	9	36
State representation	9	15	2	0	10	36
Probability rule	7	6	12	0	11	36
Physical state selection	3	1	23	0	9	36
Generator/spectral dynamics	15	11	7	0	3	36
Evolution/well-posedness	15	6	12	0	3	36
Causal propagation/Green	3	2	21	0	10	36
Gauge/BV/cohomology	4	2	15	5	10	36
Interaction construction	8	3	3	3	19	36
Counterterm classification	1	2	10	4	19	36
Anomaly classification	1	2	10	4	19	36

Obligation	Local	Literature	Pieces	Priority gap	Not mapped	Total
Renormalized products	0	2	11	4	19	36
QME restoration	0	0	13	4	19	36
Residual quantum transfer	0	0	13	4	19	36
Reconstruction/limits	3	14	1	2	16	36
All obligations	88	93	160	30	205	576

A local or literature result remains bounded by its attached claim boundary. Pieces-only means relevant ingredients do not yet compose the target. Priority gap marks a coherent current-programme gap. Not mapped is not an absence theorem. Of the 205 not-mapped coordinates, 81 are emitted coordinates reviewed without a substantive transfer and 124 are synthetic complements used to expose the full surface.

A.3 Typed implication ledger

The implication view is directional. Table 6 is the authoritative print reading of its arrows: the relation column says how the edge may be used, and the assertion column says exactly what has and has not crossed that edge.

Table 6: Complete typed relation ledger from the implications tab.

From	Relation	To	Exact assertion	Evidence
Finite resolution	Enough for this step	Finite Laurent wave	Resolving finitely many Fourier modes is enough to construct the exact finite Laurent-wave fixture.	`FOUNDATIONAL_CYLINDER_WAVE_STRENGTH_LADDER_V1`
Supplied error control	Depends on the coding	Energy-tail modulus	A usable cutoff rule supplies a tail modulus only because that quantitative rule is part of the chosen representation.	`FOUNDATIONAL_CYLINDER_WAVE_STRENGTH_LADDER_V1`
Energy-tail modulus	Enough for this step	Coded Hilbert evolution	A prescribed fast Cauchy rate and finite-code time moduli support the explicit diagonal extension in RCA_0.	`FOUNDATIONAL_CODED_POLYGONAL_WAVE_RCA0_V1`
Finite total energy	Unproved bridge	Energy-tail modulus	Finite total energy alone does not supply a rate of tail convergence; deriving one uniformly remains an open bridge.	Open bridge—no direct certificate
Coded Hilbert evolution	Unproved bridge	Finite-test wave identity	The completed evolution name has been constructed, but verifying the wave identity against the declared finite test class remains open.	`FOUNDATIONAL_CODED_POLYGONAL_WAVE_RCA0_V1`
Finite-test wave identity	Unproved bridge	Localized spacetime distribution	An identity on finite Fourier tests has not yet been extended to a localized spacetime test-function class.	Open bridge—no direct certificate
Finite Laurent wave	This method fails	Causal Green maps	Finite spectral projection is globally nonzero, so this route cannot certify causal support.	`FOUNDATIONAL_CYLINDER_WAVE_STRENGTH_LADDER_V1`
Localized spacetime distribution	Not enough by itself	Causal Green maps	A spacetime distribution alone is not enough: energy uniqueness and support propagation are additional requirements.	`FOUNDATIONAL_TYPED_BIWAVE_GREEN_DEPENDENCY_AUDIT_V1`
Finite propagation speed	Unproved bridge	Causal Green maps	Finite propagation is the physical requirement, but an explicit theorem must still construct advanced and retarded maps with controlled support.	`FOUNDATIONAL_TYPED_BIWAVE_GREEN_DEPENDENCY_AUDIT_V1`

From	Relation	To	Exact assertion	Evidence
Computable wave propagation	Literature contrast	Noncomputable solution value	Computability changes with topology, regularity, and representation; the two literature results are not contradictory.	`weihrauch-zhong-2002`; `pour-el-richards-1981`

A.4 Cylinder-wave strength ladder

The ladder prevents a certified lower level from being read as a theorem at a stronger level. "Adds" names the new mathematical data at the gate; "establishes" records the result available there; and "still open or excluded" blocks silent promotion.

Table 7: All six gates in the cylinder-wave strength ladder.

Gate, object, and sufficient base	Adds	Establishes	Still open or excluded
`L0_FINITE_LAURENT` / `CERTIFIED`. A labelled finite family of rational-complex chiral modes. Base: PRA for every fixed fixture.	finite sums; integer differentiation weights; decidable rational equality	exact wave-equation residual zero; Galerkin nesting; exact positive finite energy	completion; spatial localization; causal support
`L1_NAMED_TAIL_MODULUS` / `CERTIFIED`. The explicit datum c_n=1/n^2 together with N(k)=2^k. Base: Primitive-recursive rational inequalities plus induction for the displayed telescoping bound.	a tolerance-to-cutoff function supplied as data	a constructive Cauchy name for this named energy datum	a uniform modulus extractor for every classically convergent energy series
`L2_CODED_ENERGY_CARRIER` / `CERTIFIED`. Fast-Cauchy completion of mean-zero rational polygonal chiral pairs. Base: RCA_0 for the declared representation.	coded Cauchy completion with prescribed rate; exact rational translation group; finite-code time moduli; primitive-recursive diagonal for named real times	completed energy state; energy-conserving real-time solution name; Cauchy existence and uniqueness in the declared carrier	necessity or reversal; representation invariance; localized weak equation; causal support
`L3_COEFFICIENT_WEAK_SOLUTION` / `FORMALIZATION_TARGET`. An energy solution tested against finite Fourier sequences. Base: Not classified. This coefficient-weak notion avoids claiming a spacetime distribution or localized support.	finite-support test-sequence pairing; termwise weak equation	—	coded localized test class; coefficient-to-distribution comparison; weakest base; L2 supplies a completed evolution name, but the present certificate does not formalize test-function integration.
`L4_SPACETIME_DISTRIBUTION` / `OPEN`. A distribution on R x S^1 tested against localized smooth functions. Base: Not classified.	coded test-function space; integration or Fourier/test-function comparison; continuity in the test-function topology	—	Neither finite Laurent algebra nor a Hilbert coefficient sequence supplies this automatically.
`L5_CAUSAL_GREEN_OPERATOR` / `CONDITIONAL_IMPORT_ONLY`. Advanced and retarded solution maps with finite propagation and uniqueness. Base: Not classified.	localized energy estimates; Cauchy uniqueness; support propagation; slab globalization; distributional duality	—	The imported biwave result is conditional and its weakest base and Choice strength remain open.

A.5 Complete evidence and literature registers

The normalized catalogue contains 69 evidence records: 18 local result records and 51 literature records. Tables 8–10 print the complete catalogue rather than only the records highlighted by the graph and ladder. Literature entries preserve their artifact status: metadata-only records are bibliographic leads, not content-pinned evidence. A catalogue entry supports only its stated role and never silently crosses its recorded boundary.

Catalogue lifecycle or artifact status	Records
`CONTENT_PINNED`	39
`METADATA_ONLY`	12
`SEPARATED`	6
`SUFFICIENCY_PROVED`	5
`CERTIFIED`	2
`L2_UPPER_BOUND_CERTIFIED`	2
`LITERATURE_SCOPED`	2
`VERIFIED_BOUNDED_MODEL`	1

A.5.1 Complete literature reference list

The stable link is the catalogue's preferred public locator. `CONTENT_PINNED` means that the reviewed artifact is content-addressed in the evidence ledger; `METADATA_ONLY` means that the bibliographic identity is resolved but the source content is not pinned. The latter is a research lead, not equivalent evidential weight.

Table 8: Complete literature reference and evidence register (51 records).

Reference and provenance	Recorded evidential role and boundary
`abramsky-coecke-2004`. Samson Abramsky and Bob Coecke, A categorical semantics of quantum protocols, 2004. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/quant-ph/0402130	*Supports:* Compact closed categories with biproducts express finite quantum protocols compositionally, with scalars and a Born-rule-like probabilistic interpretation emerging from the categorical structure. *Boundary:* Categorical semantics is not itself a selected topos, a continuum field carrier, a constructive metatheory, or a derivation of physical Hilbert space from Weyl gravity.
`baer-2015`. Christian Bär, Green-hyperbolic operators on globally hyperbolic spacetimes, Communications in Mathematical Physics 333 (2015), 1585-1615. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/1310.0738	*Supports:* Normally hyperbolic wave operators are Green-hyperbolic on globally hyperbolic spacetimes; advanced and retarded maps have the declared support and extend continuously to several support classes.; For symmetric hyperbolic systems the paper proves Cauchy uniqueness, existence, finite propagation and continuous dependence. *Boundary:* This is a classical smooth/distributional theorem. It does not code the proof in second-order arithmetic, eliminate Choice, give a Bishop-constructive proof, or construct the Weyl metric BV propagator.
`bahr-dittrich-2009`. Benjamin Bahr and Bianca Dittrich, Breaking and restoring of diffeomorphism symmetry in discrete gravity, 2009. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/0909.5688	*Supports:* Discretization can break diffeomorphism symmetry and coarse-graining or perfect-action methods can be used to study its restoration. *Boundary:* The analysis does not give a Weyl-BV cohomology certificate, establish quantum anomaly cancellation, or prove a continuum limit for this repository.
`barnich-brandt-henneaux-2000`. Glenn Barnich, Friedemann Brandt, and Marc Henneaux, Local BRST cohomology in gauge theories, Physics Reports 338 (2000), 439-569. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/hep-th/0002245	*Supports:* Local BRST cohomology classifies the anomaly consistency condition, counterterms, and classical deformations for broad classes of gauge theories. *Boundary:* This is a classical-standard local cohomology framework. It does not compute Weyl-gravity coefficients, restore this repository's QME, or calibrate weak, constructive, or choice-free foundations.

Reference and provenance	Recorded evidential role and boundary
`bateman-turok-2026`. Sam Bateman and Neil Turok, Escape from Ostrogradsky via Hidden Ghost Parity (2026). `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/2607.00096	*Supports:* The Bateman-Turok proposal changes the state-space metric and adjoint/probability structure by using a Krein carrier and a fundamental ghost parity. *Boundary:* The paper's tree-level positivity and deferred operator claims retain the boundaries audited in the repository; no foundational-strength result follows from this citation.
`bender-boettcher-1998`. Carl M. Bender and Stefan Boettcher, Real spectra in non-Hermitian Hamiltonians having PT symmetry, Physical Review Letters 80 (1998), 5243-5246. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/physics/9712001	*Supports:* Families of non-Hermitian PT-symmetric Hamiltonians can have real positive spectra in an unbroken-symmetry regime. *Boundary:* PT symmetry is not identical to Krein self-adjointness, and real spectrum alone does not supply a positive physical inner product, unitary dynamics, or a gauge-QFT state space.
`blackadar-farah-2026`. Bruce Blackadar and Ilijas Farah, Separable C*-algebras Without the Countable Axiom of Choice (2026). `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/2602.15812	*Supports:* A substantial theory of separable C*-algebras, including representation, polynomial spectral mapping, and continuous functional calculus for commuting normal elements, can be developed in ZF.; Without Choice, nonseparable examples can break familiar state-space and commutative representation conclusions. *Boundary:* Does not automatically remove Choice from AQFT, interacting QFT, state selection, or nonseparable operator-algebra arguments.
`blackadar-farah-karagila-2026`. Bruce Blackadar, Ilijas Farah, and Asaf Karagila, Hilbert Spaces Without The Countable Axiom of Choice, revised 2026. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/2304.09602	*Supports:* Hilbert spaces and their operator theory can be studied in ZF without countable Choice, and pathologies expose distinctions hidden by ZFC.; The assertion that every Hilbert space has the familiar basis behaviour is not a definition-level truth independent of set theory. *Boundary:* Foundational Hilbert-space analysis, not evidence that a particular physical Hilbert space exhibits the pathological examples.
`brattka-2008`. Vasco Brattka, How incomputable is the separable Hahn-Banach theorem? (2008). `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/0808.1663	*Supports:* The separable Hahn-Banach theorem also admits a uniform computability analysis, connecting its reverse-mathematical WKL calibration to computable analysis. *Boundary:* A Weihrauch or computability classification is not automatically the same relation as implication over RCA_0 or derivability in constructive set theory.
`brenna-flori-2012`. W. Brenna and Cecilia Flori, Complex Numbers, One-Parameter of Unitary Transformations and Stone's Theorem in Topos Quantum Theory, 2012. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/1206.0809	*Supports:* The topos approach admits an internal treatment of one-parameter groups and a counterpart of Stone's theorem. *Boundary:* An internal Stone theorem is not spacetime propagation, Green-hyperbolicity, renormalization, or an empirical equivalence theorem.
`bridges-svozil-2000`. Douglas Bridges and Karl Svozil, Constructive Mathematics and Quantum Physics, International Journal of Theoretical Physics 39 (2000), 503-515. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://doi.org/10.1023/A:1003613131948	*Supports:* Bishop-style constructive mathematics can isolate constructive properties of Hilbert subspaces and projections and formulate a constructive quantum-logical axiom system. *Boundary:* The paper is an attempt at constructive quantum foundations, not a complete constructive dynamics, continuum QFT, or gravity construction.
`bridges-wang-1998-dirichlet`. Douglas Bridges and Luminiţa Vîţă Wang, Constructive aspects of the Dirichlet problem, Bulletin of the London Mathematical Society 30 (1998), 579-588. `PRIMARY_RESEARCH / METADATA_ONLY`. https://doi.org/10.1112/S0024610798006243	*Supports:* The paper constructively treats weak solutions of the Dirichlet problem and dependence on boundary or domain data in its elliptic setting. *Boundary:* This is genuine constructive PDE evidence, but it is elliptic rather than hyperbolic and supplies no causal Green maps.
`brown-simpson-1986`. Douglas K. Brown and Stephen G. Simpson, Which set existence axioms are needed to prove the separable Hahn-Banach theorem?, Annals of Pure and Applied Logic 31 (1986), 123-144. `PRIMARY_RESEARCH / METADATA_ONLY`. https://doi.org/10.1016/0168-0072(86)90066-7	*Supports:* Over RCA_0, the separable Hahn-Banach theorem studied in the paper is equivalent to WKL_0. *Boundary:* This is a reverse-mathematical result for a coded separable theorem, not the ZF choice strength of unrestricted Hahn-Banach and not proof that a concrete physics calculation invokes it.

Reference and provenance	Recorded evidential role and boundary
`brunetti-fredenhagen-rejzner-2013`. Romeo Brunetti, Klaus Fredenhagen, and Katarzyna Rejzner, Quantum gravity from the point of view of locally covariant quantum field theory, Communications in Mathematical Physics 345 (2016), 741-779. `PRIMARY_RESEARCH / CONTENT_PINNED`. `https://arxiv.org/abs/1306.1058`	*Supports:* Perturbative quantum gravity can be formulated as an effective locally covariant theory using renormalized BV methods and relational observables. *Boundary:* This does not establish perturbative renormalizability, the Weyl-gravity QME, a nonperturbative theory, or a weak/constructive foundational calibration.
`brunetti-fredenhagen-verch-2001`. Romeo Brunetti, Klaus Fredenhagen, and Rainer Verch, The generally covariant locality principle: A new paradigm for local quantum field theory, Communications in Mathematical Physics 237 (2003), 31-68. `PRIMARY_RESEARCH / CONTENT_PINNED`. `https://arxiv.org/abs/math-ph/0112041`	*Supports:* Locally covariant QFT is formulated as a functor from globally hyperbolic spacetimes to star-algebras, with admissible state spaces and relative Cauchy evolution. *Boundary:* The framework supplies architecture, not a Weyl-gravity model, a preferred state, a weak-foundation audit, or the missing full metric-BV Hadamard construction.
`chiribella-dariano-perinotti-2011`. Giulio Chiribella, Giacomo Mauro D'Ariano, and Paolo Perinotti, Informational derivation of quantum theory, Physical Review A 84, 012311 (2011). `PRIMARY_RESEARCH / CONTENT_PINNED`. `https://doi.org/10.1103/PhysRevA.84.012311`	*Supports:* Within an operational-probabilistic framework, five informational axioms define a broad class and purification selects quantum theory without taking Hilbert space as the starting postulate. *Boundary:* A reconstruction result in its declared operational framework; its proof has not yet been audited for logical or choice strength here.
`constantin-doring-2020`. Carmen Maria Constantin and Andreas Doring, A topos theoretic notion of entropy, 2020. `PRIMARY_RESEARCH / CONTENT_PINNED`. `https://arxiv.org/abs/2006.03139`	*Supports:* For finite-dimensional systems of dimension at least three, contextual entropy determines the density matrix and the paper gives an explicit reconstruction algorithm. *Boundary:* This is finite-dimensional mathematical state reconstruction, not physical state selection, field dynamics, or an internal renormalized gauge theory.
`coquand-spitters-2009`. Thierry Coquand and Bas Spitters, Constructive Gelfand duality for C*-algebras, Math. Proc. Cambridge Philos. Soc. 147 (2009), 339-344. `PRIMARY_RESEARCH / CONTENT_PINNED`. `https://doi.org/10.1017/S0305004109002515`	*Supports:* Commutative Gelfand duality admits a constructive formulation in which spectra are handled point-free. *Boundary:* Constructive commutative duality is not a complete constructive formulation of interacting quantum field theory.
`dittrich-2012`. Bianca Dittrich, From the discrete to the continuous: Towards a cylindrically consistent dynamics, New Journal of Physics 14 (2012), 123004. `PRIMARY_RESEARCH / CONTENT_PINNED`. `https://arxiv.org/abs/1205.6127`	*Supports:* Cylindrical consistency, embedding maps, and coarse graining provide explicit obligations for connecting discrete dynamics to a continuum theory. *Boundary:* This is a continuum-construction framework, not a completed continuum theorem for Weyl gravity or a foundational rejection of actual infinity.
`doring-2008`. Andreas Doring, Quantum states and measures on the spectral presheaf, 2008. `PRIMARY_RESEARCH / CONTENT_PINNED`. `https://arxiv.org/abs/0809.4847`	*Supports:* Normal states on a von Neumann algebra are related to measures on the spectral presheaf in the topos approach to quantum theory. *Boundary:* A state-measure representation is not a physical state-selection rule, a Born-rule derivation, or an internal interacting field theory.
`esmeral-ferrer-wagner-2015`. K. Esmeral, O. Ferrer, and E. Wagner, Frames in Krein spaces arising from a non-regular W-metric, Banach Journal of Mathematical Analysis 9 (2015), 1-16. `PRIMARY_RESEARCH / CONTENT_PINNED`. `https://doi.org/10.15352/bjma/09-1-1`	*Supports:* A Krein space has an indefinite product together with a fundamental symmetry whose associated positive product supplies a Hilbert-space topology. *Boundary:* A mathematical structural fact; it neither validates a generalized Born rule nor removes set-theoretic assumptions from infinite-dimensional analysis.
`fewster-verch-2011`. Christopher J. Fewster and Rainer Verch, Dynamical locality of the free scalar field, Annales Henri Poincare 13 (2012), 1675-1709. `PRIMARY_RESEARCH / CONTENT_PINNED`. `https://arxiv.org/abs/1109.6732`	*Supports:* Dynamical locality compares kinematic and dynamically defined local content and is established for classical and quantized scalar models, with explicit exceptional cases. *Boundary:* The scalar examples are not a theorem for gauge theories or Weyl gravity; the paper itself identifies gauge-theoretic complications.

Reference and provenance	Recorded evidential role and boundary
fredenhagen-rejzner-2011. Klaus Fredenhagen and Katarzyna Rejzner, Batalin-Vilkovisky formalism in perturbative algebraic quantum field theory, Communications in Mathematical Physics 317 (2013), 697-725. PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/1110.5232	*Supports:* The BV formalism can be combined with perturbative AQFT, including renormalized time-ordered products and the anomalous master Ward identity. *Boundary:* This does not instantiate the full Weyl metric complex or certify its Lorentzian QME, and it contains no reverse-mathematical or constructive strength analysis.
gibbons-hoffman-wootters-2004. Kathleen S. Gibbons, Matthew J. Hoffman, and William K. Wootters, Discrete phase space based on finite fields, Physical Review A 70, 062101 (2004). PRIMARY_RESEARCH / CONTENT_PINNED. https://doi.org/10.1103/PhysRevA.70.062101	*Supports:* For finite-dimensional quantum systems of prime-power dimension, finite-field phase space supports discrete Wigner functions and mutually unbiased line-associated bases. *Boundary:* A finite kinematics for selected dimensions is not a finite replacement for continuum dynamics, Lorentzian QFT, or a convergence theorem.
gottschalk-2004. Hanno Gottschalk, Complex velocity transformations and the Bisognano-Wichmann theorem for quantum fields acting on Krein spaces, 2004. PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/math-ph/0408048	*Supports:* Relativistic local fields on Krein space admit analytic-vector and modular-theoretic results under stated axioms, extending a Bisognano-Wichmann-type characterization. *Boundary:* This is an indefinite-metric QFT result under strong axioms, not a construction of the full Weyl metric BV theory, positive Born probabilities, or choice-free analysis.
grinkevich-1996. Yuri B. Grinkevich, Synthetic Differential Geometry: A Way to Intuitionistic Models of General Relativity in Toposes (1996). PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/gr-qc/9608013	*Supports:* Synthetic differential geometry in a suitable topos supports intuitionistic models of Riemannian geometry and Einstein equations on formal manifolds. *Boundary:* This is a classical-gravity reformulation route; it does not supply a constructive quantum Weyl theory or an external equivalence theorem for every classical spacetime.
haag-kastler-1964. Rudolf Haag and Daniel Kastler, An Algebraic Approach to Quantum Field Theory, Journal of Mathematical Physics 5 (1964), 848-861. PRIMARY_RESEARCH / METADATA_ONLY. https://doi.org/10.1063/1.1704187	*Supports:* Local quantum field theory can be axiomatized algebra-first in terms of observables and locality rather than beginning with a preferred global Hilbert-space realization. *Boundary:* Algebra-first formulation moves but does not erase representation, state-existence, topology, infinity, or set-existence questions.
harding-heunen-2019. John Harding and Chris Heunen, Topos quantum theory with short posets, 2019. PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/1903.01897	*Supports:* A smaller context poset yields a different topos while retaining core results on Kochen-Specker obstruction, spectral representation, state measures, and dynamics. *Boundary:* Changing the context poset changes internal logic; preservation of these core results is not a general invariance theorem for field theory, BV, or empirical predictions.
hardy-2001. Lucien Hardy, Quantum Theory From Five Reasonable Axioms (2001). PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/quant-ph/0101012	*Supports:* Finite-dimensional quantum formalism can be reconstructed from operational axioms, with continuity doing explicit work in separating the quantum and classical cases in this framework. *Boundary:* Finite-dimensional reconstruction under its stated framework; it does not establish the foundations of infinite-dimensional QFT or the set-theoretic strength of the reconstruction proof.
henry-2014. Simon Henry, Constructive Gelfand duality for non-unital commutative C*-algebras (2014). PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/1412.2009	*Supports:* Constructive, localic Gelfand duality extends to non-unital commutative C*-algebras and locally compact locales. *Boundary:* Commutative localic duality does not by itself construct noncommutative interacting QFT or its states and dynamics.
heunen-landsman-spitters-2009. Chris Heunen, Nicolaas P. Landsman, and Bas Spitters, A topos for algebraic quantum theory, Communications in Mathematical Physics 291 (2009), 63-110. PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/0709.4364	*Supports:* An algebraic quantum system can be represented internally in a topos with an intuitionistic Heyting logic and a localic spectrum. *Boundary:* A reformulation of algebraic quantum theory; it does not establish empirical superiority, eliminate all external classical reasoning, or construct Weyl QFT.

Reference and provenance	Recorded evidential role and boundary
`humphreys-simpson-1996`. A. James Humphreys and Stephen G. Simpson, Separable Banach space theory needs strong set existence axioms, Transactions of the American Mathematical Society 348 (1996), 4231-4255. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://doi.org/10.1090/S0002-9947-96-01742-6	*Supports:* Some weak-star closure existence statements for separable Banach-space duals require Pi^1_1 comprehension, so separability alone does not guarantee uniformly weak foundational strength. *Boundary:* This does not make every theorem about a separable Banach space strong; the coded target and its representation determine the reversal.
`humphreys-simpson-1999`. A. James Humphreys and Stephen G. Simpson, Separation and Weak Konig's Lemma, Journal of Symbolic Logic 64 (1999), 268-278. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://doi.org/10.2307/2586763	*Supports:* Over RCA_0, separation for open convex sets is equivalent to WKL_0, while separation for separably closed convex sets is equivalent to ACA_0.; The representation of a closed convex set changes the logical strength of an apparently similar separation claim. *Boundary:* The result calibrates two precise separable Banach-space statements; it does not assign one strength to every use of geometric separation.
`kogut-susskind-1975`. John Kogut and Leonard Susskind, Hamiltonian formulation of Wilson's lattice gauge theories, Physical Review D 11 (1975), 395-408. `PRIMARY_RESEARCH / METADATA_ONLY`. https://doi.org/10.1103/PhysRevD.11.395	*Supports:* Wilson lattice gauge theory admits a Hamiltonian formulation with gauge constraints and a continuum-limit programme. *Boundary:* The historical formulation is not finite in every mathematical sense and does not supply a controlled Weyl-gravity continuum or BV-QME certificate.
`kostrykin-potthoff-schrader-2011`. Vadim Kostrykin, Jürgen Potthoff, and Robert Schrader, Finite propagation speed for solutions of the wave equation on metric graphs, 2011. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/1106.0817	*Supports:* A class of self-adjoint Laplace operators on metric graphs has existence and uniqueness for the wave equation and strict finite propagation, proved by localized energy methods. *Boundary:* Metric graphs retain continuous edges and Hilbert/Sobolev analysis. They are not finite exact algebra, a continuum-limit theorem, or a choice-free construction.
`mostafazadeh-2001`. Ali Mostafazadeh, Pseudo-Hermiticity versus PT symmetry: The necessary condition for the reality of the spectrum of a non-Hermitian Hamiltonian, Journal of Mathematical Physics 43 (2002), 205-214. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/math-ph/0107001	*Supports:* Pseudo-Hermiticity is introduced and shown to be a necessary structural condition for Hamiltonians with real spectrum in the setting studied. *Boundary:* The result assumes analytic and spectral hypotheses and does not by itself define physical probabilities, interacting QFT, or a weakest foundational base.
`muehlhoff-2010`. Rainer Mühlhoff, Cauchy Problem and Green's Functions for First Order Differential Operators and Algebraic Quantization, Journal of Mathematical Physics 52 (2011), 022303. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/1001.4091	*Supports:* Prenormally hyperbolic first-order operators have unique advanced and retarded Green functions and a globally well-posed Cauchy problem under the stated globally hyperbolic hypotheses. *Boundary:* The reduction imports the normally-hyperbolic second-order theorem and remains classical; it is not a foundational-strength or Weyl-BV result.
`nachtergaele-raz-schlein-sims-2007`. Bruno Nachtergaele, Hillel Raz, Benjamin Schlein, and Robert Sims, Lieb-Robinson Bounds for Harmonic and Anharmonic Lattice Systems, Communications in Mathematical Physics 286 (2009), 1073-1098. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/0712.3820	*Supports:* Harmonic and specified anharmonic lattice systems satisfy Lieb-Robinson bounds, including exponentially small commutators outside an effective cone for Weyl observables. *Boundary:* An exponentially small tail is not strict support and is not an advanced/retarded Green operator. The result must not be promoted to continuum Lorentzian causality.
`neumann-pape-streicher-2018`. Eike Neumann, Martin Pape, and Thomas Streicher, Computability in Basic Quantum Mechanics, Logical Methods in Computer Science 14(2:14) (2018), 1-20. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/1610.09209	*Supports:* For separable infinite-dimensional Hilbert spaces, states and observables can be represented by measures and valuations so that an effective spectral-theorem analysis becomes possible. *Boundary:* Computable representations are not equivalent to a Bishop-constructive proof or an RCA_0 reversal, and the result does not supply interacting QFT.
`pischke-2025-semigroups`. Nicholas Pischke, A proof-theoretic metatheorem for nonlinear semigroups generated by an accretive operator and applications, Selecta Mathematica 31 (2025), 32. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/2304.01723	*Supports:* Logical metatheorems cover nonlinear semigroups generated by accretive operators and support extraction of quantitative convergence information in the stated systems and case studies. *Boundary:* Proof mining in higher-type systems is not an RCA_0/WKL_0/ACA_0 reversal, and the operator and semigroup structures are encoded assumptions rather than a causal PDE construction.

Reference and provenance	Recorded evidential role and boundary
`pour-el-richards-1981`. Marian B. Pour-El and J. Ian Richards, The wave equation with computable initial data such that its unique solution is not computable, Advances in Mathematics 39 (1981), 215-239. `PRIMARY_RESEARCH / METADATA_ONLY`. https://doi.org/10.1016/0001-8708(81)90001-3	*Supports:* Under the paper's representations, computable initial data can evolve to a unique wave solution with noncomputable values. *Boundary:* The result is representation-sensitive and is not a no-go theorem for computable Sobolev wave propagation or for the explicit coefficient carrier used here.
`richman-bridges-1999`. Fred Richman and Douglas Bridges, A Constructive Proof of Gleason's Theorem, Journal of Functional Analysis 162 (1999), 287-312. `PRIMARY_RESEARCH / METADATA_ONLY`. https://doi.org/10.1006/jfan.1998.3372	*Supports:* Gleason's theorem has a constructive proof after its hypotheses and conclusion are given an appropriate constructive formulation. *Boundary:* This is a reformulation and proof of a specific probability-representation theorem; it is not a constructive derivation of all quantum mechanics.
`selivanova-selivanov-2013`. Svetlana Selivanova and Victor Selivanov, Computing Solution Operators of Boundary-value Problems for Some Linear Hyperbolic Systems of PDEs, Logical Methods in Computer Science 13(4:13) (2017). `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/1305.2494	*Supports:* For symmetric hyperbolic systems on a cube with computable coefficients, the Cauchy solution operator is computable in the stated TTE representations; dissipative boundary-value problems are also treated under additional hypotheses.; The proof uses rational finite-difference approximants with effective error estimates rather than an explicit solution formula. *Boundary:* A TTE computability theorem is not a Bishop-constructive derivation, an RCA_0 upper bound or reversal, or a theorem for globally hyperbolic manifolds and advanced/retarded Green support.
`selivanova-selivanov-2018`. Svetlana Selivanova and Victor Selivanov, Bit Complexity of Computing Solutions for Symmetric Hyperbolic Systems of PDEs with Guaranteed Precision, 2020. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/1807.03140	*Supports:* The symmetric-hyperbolic computability programme admits explicit bit-complexity upper bounds under the paper's representations and coefficient hypotheses. *Boundary:* Complexity of represented solution operators neither supplies strict causal Green support nor calibrates a subsystem of second-order arithmetic.
`simpson-1984-ode`. Stephen G. Simpson, Which set existence axioms are needed to prove the Cauchy/Peano theorem for ordinary differential equations?, Journal of Symbolic Logic 49 (1984), 783-802. `PRIMARY_RESEARCH / METADATA_ONLY`. https://doi.org/10.2307/2274131	*Supports:* Over RCA_0, the Cauchy-Peano existence theorem for ordinary differential equations is equivalent to WKL_0; stronger related ODE solution principles reach ACA_0. *Boundary:* This calibrates ODE existence, not a hyperbolic PDE energy theorem, finite propagation, or a Green map.
`weihrauch-zhong-2002`. Klaus Weihrauch and Ning Zhong, Is wave propagation computable or can wave computers beat the Turing machine?, Proceedings of the London Mathematical Society 85 (2002), 312-332. `PRIMARY_RESEARCH / METADATA_ONLY`. https://doi.org/10.1112/S0024611502013643	*Supports:* The paper proves computability of the wave propagator for specified continuously differentiable and Sobolev representations, with a loss of one derivative in the former setting and no loss in the stated Sobolev setting. *Boundary:* This is TTE computable analysis, not a reverse-mathematical RCA_0/WKL_0/ACA_0 classification and not specifically the cylinder fixture.
`weihrauch-zhong-2006-fundamental`. Klaus Weihrauch and Ning Zhong, An Algorithm for Computing Fundamental Solutions, SIAM Journal on Computing 35 (2006), 1283-1294. `PRIMARY_RESEARCH / METADATA_ONLY`. https://doi.org/10.1137/S0097539704446360	*Supports:* The paper computes a distributional fundamental solution for every constant-coefficient differential operator in its represented setting. *Boundary:* A fundamental solution is not automatically a retarded or advanced Green operator and the record does not establish causal support.
`weihrauch-zhong-2007-cauchy`. Klaus Weihrauch and Ning Zhong, Computable analysis of the abstract Cauchy problem in a Banach space and its applications I, Mathematical Logic Quarterly 53 (2007), 511-531. `PRIMARY_RESEARCH / METADATA_ONLY`. https://doi.org/10.1002/malq.200710015	*Supports:* The paper gives necessary and sufficient conditions for computability of the solution operator of an abstract linear Cauchy problem with a possibly unbounded Banach-space operator in the representation approach. *Boundary:* This is TTE computability, not Bishop constructivity or reverse mathematics, and it does not prove finite propagation.
`zhong-1999-sobolev`. Ning Zhong, Computability structure of the Sobolev spaces and its applications, Theoretical Computer Science 219 (1999), 487-510. `PRIMARY_RESEARCH / METADATA_ONLY`. https://doi.org/10.1016/S0304-3975(98)00302-8	*Supports:* The paper defines computability structures on Sobolev spaces and applies them to classes of second-order hyperbolic and parabolic PDEs. *Boundary:* A computable-analysis result is not a Bishop proof, a subsystem reversal, or a causal-support theorem.

Reference and provenance	Recorded evidential role and boundary
zhong-weihrauch-2003-distributions. Ning Zhong and Klaus Weihrauch, Computability theory of generalized functions, Journal of the ACM 50 (2003), 469-505. PRIMARY_RESEARCH / METADATA_ONLY. https://doi.org/10.1145/792538.792542	*Supports:* The paper defines TTE computability for test functions and distributions and proves computability of the solution operator for the distributional inhomogeneous three-dimensional wave equation. *Boundary:* The reviewed abstract does not certify retarded or advanced selection or a strict cone-support theorem; this is therefore not direct causal-Green evidence.
zohar-burrello-2014. Erez Zohar and Michele Burrello, Formulation of lattice gauge theories for quantum simulations, Physical Review D 91, 054506 (2015). PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/1409.3085	*Supports:* Hamiltonian lattice gauge models can encode local gauge invariance and Gauss-law constraints for finite, compact, and truncated gauge groups. *Boundary:* A simulator-oriented lattice Hamiltonian is not a finite Weyl-gravity BV complex, a regulator-independent continuum limit, or a QME certificate.

A.5.2 Complete local result and certificate list

Local records are repository results, not external publications. Their positive flags state what the registered certificate establishes; the complete negative list prevents a bounded result from being promoted to a stronger mathematical or physical claim.

Table 9: Complete local result and certificate register (18 records).

Record and repository locator	Certified scope and complete boundary
FOUNDATIONAL_BT_SEPARABLE_STATE_CHAIN_ZF_V1. FOUNDATIONAL_DEPENDENCY_CERTIFICATE / SEPARATED. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: foundations/results/FOUNDATIONAL_BT_SEPARABLE_STATE_CHAIN_ZF_V1.json. Report: foundations/reports/bt-separable-cstar-state-chain.md.	*Establishes:* Certified positive flags: separable detector algebra constructed; explicit zf states constructed; explicit corner gns constructed. *Does not establish:* that B(l2(Z)) is separable; a finite trace of the identity; a canonical choice of incoming detector projection; a thermodynamic normal state; that the coherent CCR state is the BT physical state; full nonlinear Moller dynamics or Eq. (19); a gravitational or BRST lift; anything LORENTZIAN-CAUSAL.
FOUNDATIONAL_CODED_POLYGONAL_WAVE_RCA0_V1. CODED_SECOND_ORDER_ARITHMETIC_UPPER_BOUND / CERTIFIED. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: foundations/results/FOUNDATIONAL_CODED_POLYGONAL_WAVE_RCA0_V1.json. Report: foundations/reports/coded-polygonal-wave-rca0.md.	*Establishes:* Certified positive flags: rca0 upper bound for declared representation; completed energy state constructed; real time solution name constructed; energy conservation proved; cauchy uniqueness in declared carrier. *Does not establish:* that RCA_0 is necessary or the weakest base; a WKL_0, ACA_0, or Choice lower bound; the same upper bound for bare finite-energy existence without a fast Cauchy name; representation invariance; a localized spacetime-distribution theorem; finite propagation or an advanced/retarded Green map; a variable-coefficient or curved-spacetime Cauchy theorem; the biwave or metric-BV propagator; a new LORENTZIAN-CAUSAL result.
FOUNDATIONAL_CODED_WAVE_FRONTIER_V2. CODED_UPPER_BOUND_AND_LITERATURE_FRONTIER / L2_UPPER_BOUND_CERTIFIED. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE, LORENTZIAN-CAUSAL. Result: foundations/results/FOUNDATIONAL_CODED_WAVE_FRONTIER_V2.json. Report: foundations/reports/coded-wave-frontier-v2.md.	*Establishes:* Certified positive flags: declared representation rca0 upper bound; literature screen completed. *Does not establish:* literature completeness; that RCA_0 is the weakest base; a reverse-mathematical equivalence for wave evolution; a Bishop-constructive hyperbolic Green theorem; a ZF-without-Choice PDE theorem; that a distributional fundamental solution is retarded or advanced; strict causal support; a new Lorentzian Weyl result.
FOUNDATIONAL_CYLINDER_WAVE_STRENGTH_LADDER_V1. FOUNDATIONAL_STRENGTH_LADDER / SEPARATED. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE, LORENTZIAN-CAUSAL. Result: foundations/results/FOUNDATIONAL_CYLINDER_WAVE_STRENGTH_LADDER_V1.json. Report: foundations/reports/cylinder-wave-strength-ladder.md.	*Establishes:* Certified positive flags: finite cylinder wave exact; explicit tail modulus exact; finite spectral locality obstruction exact; typed physics to mathematics graph constructed. *Does not establish:* that PRA proves an infinite completed energy-space theorem; RCA_0 sufficiency for the selected representation; a WKL_0, ACA_0, or stronger reversal; a uniform constructive modulus extractor from bare finite-energy existence; equivalence between coefficient-weak and spacetime-distributional solutions; finite propagation or causal support from Fourier truncations; a normally-hyperbolic Green theorem; the full biwave or metric-BV propagator; renormalized Lorentzian products or a QME theorem; a new LORENTZIAN-CAUSAL result.

Record and repository locator	Certified scope and complete boundary
`FOUNDATIONAL_CYLINDER_WAVE_STRENGTH_LADDER_V2`. `FOUNDATIONAL_STRENGTH_LADDER / L2_UPPER_BOUND_CERTIFIED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE, LORENTZIAN-CAUSAL. Result: `foundations/results/FOUNDATIONAL_CYLINDER_WAVE_STRENGTH_LADDER_V2.json`. Report: `foundations/reports/cylinder-wave-strength-ladder-v2.md`.	*Establishes:* Certified positive flags: finite cylinder wave exact; explicit tail modulus exact; finite spectral locality obstruction exact; typed physics to mathematics graph constructed; arbitrary energy completion formalized in rca0; declared representation rca0 upper bound. *Does not establish:* that RCA_0 is necessary or the weakest base; a WKL_0, ACA_0, or Choice reversal; an upper bound for representations lacking prescribed Cauchy rates; representation invariance; a coefficient-weak or localized spacetime-distribution theorem; finite propagation or causal support from Fourier or polygonal evolution; a normally-hyperbolic Green theorem; the full biwave or metric-BV propagator; renormalized Lorentzian products or a QME theorem; a new LORENTZIAN-CAUSAL result.
`FOUNDATIONAL_EXPLICIT_ENERGY_SPECTRAL_FRAGMENT_ZF_V1`. `FOUNDATIONAL_DEPENDENCY_CERTIFICATE / SUFFICIENCY_PROVED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_EXPLICIT_ENERGY_SPECTRAL_FRAGMENT_ZF_V1.json`. Report: `foundations/reports/explicit-energy-spectral-fragment-audit.md`.	*Establishes:* Certified positive flags: explicit energy self adjointness route classified; fock fixed energy finiteness route classified. *Does not establish:* ZF is the weakest base; a reversal or independence theorem; foundations of arbitrary self-adjoint operators; Euclidean Hessian spectral measures; determinants or traces; an interacting Hamiltonian; a LORENTZIAN-CAUSAL result.
`FOUNDATIONAL_EXPLICIT_MODE_DYNAMICS_ZF_V1`. `FOUNDATIONAL_DEPENDENCY_CERTIFICATE / SUFFICIENCY_PROVED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_EXPLICIT_MODE_DYNAMICS_ZF_V1.json`. Report: `foundations/reports/explicit-mode-dynamics-zf.md`.	*Establishes:* Certified positive flags: explicit strongly continuous unitary group constructed; explicit j unitary group constructed; explicit point norm cstar dynamics constructed; explicit fock unitary group constructed. *Does not establish:* that ZF is the weakest sufficient base; exponentiation of the other unbounded conformal generators; a representation of the full conformal group; nonlinear Bateman-Turok Moller or full-orbit dynamics; an interacting Weyl Hamiltonian or automorphism group; causal Green propagation or a Lorentzian off-shell BV propagator; state selection, a KMS state, or a generalized Born rule; thermodynamic-limit implementability in an inequivalent representation; anything LORENTZIAN-CAUSAL.
`FOUNDATIONAL_FINITE_FIELD_FINITE_MODE_NON_EQUIVALENCE_V1`. `FOUNDATIONAL_NON_EQUIVALENCE_LEDGER / SEPARATED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_FINITE_FIELD_FINITE_MODE_NON_EQUIVALENCE_V1.json`. Report: `foundations/reports/finite-field-versus-finite-mode.md`.	*Establishes:* Certified positive flags: four objects typed; pairwise non equivalence witnessed. *Does not establish:* a no-go theorem for finite-field physics; a continuum limit of finite-field Wigner systems; equivalence of varying prime-power dimensions; convergence of Weyl mode cutoffs; a finitist construction of real or complex analysis; empirical evidence for or against actual infinity; a finite formulation of interacting Weyl gravity; anything LORENTZIAN-CAUSAL.
`FOUNDATIONAL_FINITE_GRAPH_WAVE_CAUSALITY_V1`. `EXACT_FINITE_DISCRETE_CAUSAL_GREEN_KERNEL / CERTIFIED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_FINITE_GRAPH_WAVE_CAUSALITY_V1.json`. Report: `foundations/reports/finite-graph-wave-causality.md`.	*Establishes:* Certified positive flags: finite exact retarded kernel constructed; finite exact advanced kernel constructed; graph step support certified. *Does not establish:* continuum finite propagation; a Lorentzian advanced or retarded Green operator; CFL stability or convergence under refinement; a regulator-independent continuum limit; a Weyl metric BV propagator; a reverse-mathematical classification of continuum PDE.
`FOUNDATIONAL_FINITE_QUBIT_INTERACTION_CORE_V1`. `EXACT_FINITE_INTERACTION_WITNESS / VERIFIED_BOUNDED_MODEL`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_FINITE_QUBIT_INTERACTION_CORE_V1.json`. Report: `foundations/reports/finite-qubit-interaction-core.md`.	*Establishes:* Certified positive flags: exact finite interaction model; exact state and probability witness; exact entanglement generation witness; exact finite krein companion. *Does not establish:* a weakest logical or set-theoretic base; that PRA, ZF without Choice, Bishop constructivism, and finitism are equivalent regimes; an infinite-dimensional Hilbert, Krein, or C*-completion; a continuum field theory or controlled continuum limit; a Weyl-gravity interaction vertex; counterterm or anomaly classification; renormalization or restoration of the quantum master equation; a physical selection principle for the displayed states; a Lorentzian propagator, Hadamard state, or causal quantum theory.

Record and repository locator	Certified scope and complete boundary
`FOUNDATIONAL_FREE_BV_ENERGY2_PRA_SDR_V1`. `FOUNDATIONAL_DEPENDENCY_CERTIFICATE / SUFFICIENCY_PROVED`. Dependency tags: LOCAL-ALGEBRAIC. Result: `foundations/results/FOUNDATIONAL_FREE_BV_ENERGY2_PRA_SDR_V1.json`. Report: `foundations/reports/free-bv-energy2-pra-sdr.md`.	*Establishes:* Certified positive flags: fixed energy2 integer sdr verified; pra sufficiency for fixed checker; hahn banach avoided for displayed certificate. *Does not establish:* That PRA is the weakest possible base theory.; A reversal or necessity theorem for any comprehension or choice principle.; That the source producer, all energy levels, the Fock lift, or the complete field-derived BV complex is independently verified.; That arbitrary finite-dimensional linear algebra avoids Choice without an explicit presentation and witness.; A choice-free countable or infinite-dimensional Hilbert, Krein, spectral, Green-operator, state, or renormalization construction.; A complete classical import freeze or any publishable quantum promotion.; Any EUCLIDEAN-SPECTRAL, REDUCED-MODE, or LORENTZIAN-CAUSAL claim..
`FOUNDATIONAL_HARDY_CONTINUITY_KN_AUDIT_V1`. `FOUNDATIONAL_PROOF_DEPENDENCY_AUDIT / SEPARATED`. Dependency tags: LOCAL-ALGEBRAIC. Result: `foundations/results/FOUNDATIONAL_HARDY_CONTINUITY_KN_AUDIT_V1.json`. Report: `foundations/reports/hardy-continuity-kn-foundational-audit.md`.	*Establishes:* Certified positive flags: hardy kn step audited; rca0 sufficient for explicit modulus route. *Does not establish:* a weakest-base theorem; a reversal to RCA_0 or WKL_0; the strength of extracting a modulus from every pointwise-continuous code; the full K=N^r derivation; the K=N^2 quantum reconstruction; that Axiom 5 is empirically necessary; infinite-dimensional quantum theory; anything LORENTZIAN-CAUSAL.
`FOUNDATIONAL_KREIN_EXPLICIT_J_ZF_V1`. `FOUNDATIONAL_DEPENDENCY_CERTIFICATE / SUFFICIENCY_PROVED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_KREIN_EXPLICIT_J_ZF_V1.json`. Report: `foundations/reports/krein-explicit-j-zf-audit.md`.	*Establishes:* Certified positive flags: finite integral j verified; explicit mode index countable in zf; zf one particle completion sufficient; zf explicit bosonic fock sufficient. *Does not establish:* That ZF is the weakest foundation for the countable completion or Fock construction.; A reverse-mathematical equivalence or necessity theorem for Infinity, comprehension, Countable Choice, or Choice.; That an arbitrary indefinite inner-product space admits a fundamental decomposition without Choice.; That an arbitrary Hilbert space has an orthonormal basis in ZF.; Foundational classifications of the all-real-order Sobolev scale, closed generator domains, exponentiation, or the conformal group representation.; A trace-class density operator, normal state, generalized Born rule, particle-unitarity theorem, or interacting implementer.; A positive graviton Hilbert space, complete covariant BV propagator, Hadamard state, QME theorem, or any LORENTZIAN-CAUSAL claim..
`FOUNDATIONAL_KREIN_STATE_SELECTION_ZF_V1`. `FOUNDATIONAL_DEPENDENCY_CERTIFICATE / SUFFICIENCY_PROVED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_KREIN_STATE_SELECTION_ZF_V1.json`. Report: `foundations/reports/krein-state-selection-zf.md`.	*Establishes:* Certified positive flags: explicit zf krein vector states constructed; explicit rank one density witnesses constructed; explicit fock coordinate states constructed. *Does not establish:* a unique or canonical state from J; nonexistence of singular permutation-invariant states; that sign-preserving coordinate permutations are a required physical symmetry; a KMS, Hadamard, BRST-compatible, incoming, outgoing, or interacting state; a generalized Born rule or physical probability interpretation; trace-norm thermodynamic convergence of the Bateman-Turok construction; positivity with respect to the Krein adjoint where Hilbert-adjoint positivity was used; a weakest-base reverse-mathematics theorem; anything LORENTZIAN-CAUSAL.
`FOUNDATIONAL_LOW_HANGING_CELL_CLOSURE_AUDIT_V1`. `FOUNDATIONAL_SCOPE_AUDIT / SEPARATED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_LOW_HANGING_CELL_CLOSURE_AUDIT_V1.json`. Report: `foundations/reports/low-hanging-cell-closure-audit.md`.	*Establishes:* Certified positive flags: three existing result closures identified; local bv cohomology imported with scope; local counterterm anomaly classes imported with scope; finite cutoff dynamics imported with scope; remaining assessed open cells exhaustively triaged. *Does not establish:* a complete global smooth or distributional off-shell BV complex; the classical import freeze gate; a complete classification beyond the regular Bach locus and declared derivative bounds; counterterm or anomaly coefficients; regulated Slavnov breaking, renormalized products, or QME restoration; residual transfer or a one-particle interpretation of residual classes; a finite-to-continuum comparison or regulator-independent limit; causal propagation or any LORENTZIAN-CAUSAL theorem; that any remaining open cell is impossible or unimportant; that any of the 157 NOT_MAPPED cells has been assessed.

Record and repository locator	Certified scope and complete boundary
`FOUNDATIONAL_NORMAL_HYPERBOLIC_FACTOR_ATLAS_V1`. `LITERATURE_AND_FOUNDATIONAL_DEPENDENCY_ATLAS` / `LITERATURE_SCOPED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE, LORENTZIAN-CAUSAL. Result: `foundations/results/FOUNDATIONAL_NORMAL_HYPERBOLIC_FACTOR_ATLAS_V1.json`. Report: `foundations/reports/normal-hyperbolic-factor-foundations.md`.	*Establishes:* Certified positive flags: classical factor theorem identified; computable upper bound identified; finite exact support constructed. *Does not establish:* literature completeness; a weakest subsystem; an RCA_0, WKL_0 or ACA_0 equivalence; a Bishop-constructive globally hyperbolic Green theorem; Choice avoidance for Sobolev/distribution theory; a continuum limit from finite graphs; a full off-shell Weyl metric BV propagator; a BRST-compatible Hadamard state; renormalized Lorentzian products or a Lorentzian QME.
`FOUNDATIONAL_TOPOS_WEYL_BV_OBSTRUCTION_LEDGER_V0`. `FOUNDATIONAL_GLOSSARY_AND_OBSTRUCTION_LEDGER` / `LITERATURE_SCOPED`. Dependency tags: LOCAL-ALGEBRAIC. Result: `foundations/results/FOUNDATIONAL_TOPOS_WEYL_BV_OBSTRUCTION_LEDGER_V0.json`. Report: `foundations/reports/topos-weyl-bv-obstruction-ledger.md`.	*Establishes:* Certified positive flags: glossary complete for first artifact; obstruction dag closed. *Does not establish:* a formalized topos-internal Weyl-gravity BV complex; that every ordinary Weyl-gravity construction internalizes in an arbitrary topos; a constructive theory of distributions or wavefront sets; internal retarded or advanced Green operators; a topos-internal Krein or Hilbert completion; a physical state-selection or Born-rule theorem; renormalized time-ordered products; restoration of the quantum master equation; equivalence with the repository's external classical or quantum theory; anything LORENTZIAN-CAUSAL.
`FOUNDATIONAL_TYPED_BIWAVE_GREEN_DEPENDENCY_AUDIT_V1`. `FOUNDATIONAL_PROOF_DEPENDENCY_AUDIT` / `SEPARATED`. Dependency tags: LOCAL-ALGEBRAIC, LORENTZIAN-CAUSAL. Result: `foundations/results/FOUNDATIONAL_TYPED_BIWAVE_GREEN_DEPENDENCY_AUDIT_V1.json`. Report: `foundations/reports/typed-biwave-green-foundational-dependencies.md`.	*Establishes:* Certified positive flags: dependency cut complete for source theorem; finite resolvent algebra replayed; conditional lorentzian theorem imported. *Does not establish:* the weakest subsystem for normally-hyperbolic PDE; Choice avoidance for Sobolev or distribution theory; the energy hypotheses for an arbitrary Weyl operator; a full off-shell metric BV propagator; a BRST-compatible Hadamard state; renormalized Lorentzian time-ordered products; a Lorentzian QME theorem; quantum theory.

A.5.3 Evidence usage crosswalk

Every evidence record is used by at least one matrix coordinate. The matrix column reports the number of coordinates carrying the record and their coverage-status composition. Graph entries use edge numbers from Table 6; ladder entries use the gate identifiers from Table 7. Clicking a record identifier returns to its full literature or local-certificate entry above.

Table 10: Complete crosswalk from the atlas views to all 69 evidence records.

Evidence record	Matrix use	Graph and ladder use
`FOUNDATIONAL_BT_SEPARABLE_STATE_CHAIN_ZF_V1` (local)	11 coordinates (Local 6, Literature 1, Pieces 4).	No graph or ladder use.
`FOUNDATIONAL_CODED_POLYGONAL_WAVE_RCA0_V1` (local)	3 coordinates (Local 2, Pieces 1).	graph edges 3, 5; ladder L2_CODED_ENERGY_CARRIER.
`FOUNDATIONAL_CODED_WAVE_FRONTIER_V2` (local)	2 coordinates (Local 2).	No graph or ladder use.
`FOUNDATIONAL_CYLINDER_WAVE_STRENGTH_LADDER_V1` (local)	6 coordinates (Local 2, Literature 1, Pieces 3).	graph edges 1, 2, 7.
`FOUNDATIONAL_CYLINDER_WAVE_STRENGTH_LADDER_V2` (local)	2 coordinates (Local 2).	No graph or ladder use.
`FOUNDATIONAL_EXPLICIT_ENERGY_SPECTRAL_FRAGMENT_ZF_V1` (local)	8 coordinates (Local 6, Pieces 2).	No graph or ladder use.
`FOUNDATIONAL_EXPLICIT_MODE_DYNAMICS_ZF_V1` (local)	15 coordinates (Local 11, Pieces 4).	No graph or ladder use.
`FOUNDATIONAL_FINITE_FIELD_FINITE_MODE_NON_EQUIVALENCE_V1` (local)	2 coordinates (Local 1, Priority gap 1).	No graph or ladder use.
`FOUNDATIONAL_FINITE_GRAPH_WAVE_CAUSALITY_V1` (local)	1 coordinates (Local 1).	No graph or ladder use.
`FOUNDATIONAL_FINITE_QUBIT_INTERACTION_CORE_V1` (local)	99 coordinates (Local 45, Literature 2, Pieces 52).	No graph or ladder use.
`FOUNDATIONAL_FREE_BV_ENERGY2_PRA_SDR_V1` (local)	15 coordinates (Local 5, Pieces 10).	No graph or ladder use.

Evidence record	Matrix use	Graph and ladder use
FOUNDATIONAL_HARDY_CONTINUITY_KN_AUDIT_V1 (local)	2 coordinates (Local 2).	No graph or ladder use.
FOUNDATIONAL_KREIN_EXPLICIT_J_ZF_V1 (local)	18 coordinates (Local 11, Pieces 7).	No graph or ladder use.
FOUNDATIONAL_KREIN_STATE_SELECTION_ZF_V1 (local)	12 coordinates (Local 7, Pieces 5).	No graph or ladder use.
FOUNDATIONAL_LOW_HANGING_CELL_CLOSURE_AUDIT_V1 (local)	7 coordinates (Local 3, Pieces 4).	No graph or ladder use.
FOUNDATIONAL_NORMAL_HYPERBOLIC_FACTOR_ATLAS_V1 (local)	4 coordinates (Local 2, Pieces 2).	No graph or ladder use.
FOUNDATIONAL_TOPOS_WEYL_BV_OBSTRUCTION_LEDGER_V0 (local)	22 coordinates (Literature 2, Pieces 17, Priority gap 3).	No graph or ladder use.
FOUNDATIONAL_TYPED_BIWAVE_GREEN_DEPENDENCY_AUDIT_V1 (local)	8 coordinates (Local 4, Pieces 4).	graph edges 8, 9; ladder L5_CAUSAL_GREEN_OPERATOR.
abramsky-coecke-2004 (literature)	22 coordinates (Local 4, Literature 10, Pieces 8).	No graph or ladder use.
baer-2015 (literature)	2 coordinates (Local 2).	No graph or ladder use.
bahr-dittrich-2009 (literature)	7 coordinates (Literature 3, Pieces 4).	No graph or ladder use.
barnich-brandt-henneaux-2000 (literature)	4 coordinates (Literature 4).	No graph or ladder use.
bateman-turok-2026 (literature)	5 coordinates (Local 2, Literature 1, Pieces 2).	No graph or ladder use.
bender-boettcher-1998 (literature)	6 coordinates (Literature 2, Pieces 4).	No graph or ladder use.
blackadar-farah-2026 (literature)	9 coordinates (Local 1, Literature 4, Pieces 4).	No graph or ladder use.
blackadar-farah-karagila-2026 (literature)	6 coordinates (Literature 4, Pieces 2).	No graph or ladder use.
brattka-2008 (literature)	2 coordinates (Literature 1, Pieces 1).	No graph or ladder use.
brenna-flori-2012 (literature)	17 coordinates (Literature 10, Pieces 7).	No graph or ladder use.
bridges-svozil-2000 (literature)	2 coordinates (Literature 1, Pieces 1).	No graph or ladder use.
bridges-wang-1998-dirichlet (literature)	1 coordinates (Literature 1).	No graph or ladder use.
brown-simpson-1986 (literature)	2 coordinates (Literature 1, Pieces 1).	No graph or ladder use.
brunetti-fredenhagen-rejzner-2013 (literature)	2 coordinates (Literature 2).	No graph or ladder use.
brunetti-fredenhagen-verch-2001 (literature)	11 coordinates (Literature 7, Pieces 4).	No graph or ladder use.
chiribella-dariano-perinotti-2011 (literature)	7 coordinates (Literature 5, Pieces 2).	No graph or ladder use.
constantin-doring-2020 (literature)	14 coordinates (Local 2, Literature 7, Pieces 5).	No graph or ladder use.
coquand-spitters-2009 (literature)	14 coordinates (Literature 10, Pieces 4).	No graph or ladder use.
dittrich-2012 (literature)	7 coordinates (Literature 5, Pieces 2).	No graph or ladder use.
doring-2008 (literature)	18 coordinates (Literature 9, Pieces 9).	No graph or ladder use.
esmeral-ferrer-wagner-2015 (literature)	1 coordinates (Literature 1).	No graph or ladder use.
fewster-verch-2011 (literature)	2 coordinates (Literature 2).	No graph or ladder use.
fredenhagen-rejzner-2011 (literature)	18 coordinates (Literature 13, Pieces 5).	No graph or ladder use.
gibbons-hoffman-wootters-2004 (literature)	10 coordinates (Literature 7, Pieces 3).	No graph or ladder use.
gottschalk-2004 (literature)	13 coordinates (Literature 7, Pieces 6).	No graph or ladder use.
grinkevich-1996 (literature)	2 coordinates (Literature 2).	No graph or ladder use.
haag-kastler-1964 (literature)	5 coordinates (Local 3, Literature 1, Pieces 1).	No graph or ladder use.
harding-heunen-2019 (literature)	39 coordinates (Literature 21, Pieces 18).	No graph or ladder use.
hardy-2001 (literature)	7 coordinates (Literature 5, Pieces 2).	No graph or ladder use.
henry-2014 (literature)	2 coordinates (Literature 2).	No graph or ladder use.
heunen-landsman-spitters-2009 (literature)	33 coordinates (Literature 20, Pieces 13).	No graph or ladder use.
humphreys-simpson-1996 (literature)	1 coordinates (Literature 1).	No graph or ladder use.
humphreys-simpson-1999 (literature)	2 coordinates (Literature 1, Pieces 1).	No graph or ladder use.
kogut-susskind-1975 (literature)	11 coordinates (Literature 5, Pieces 6).	No graph or ladder use.
kostrykin-potthoff-schrader-2011 (literature)	3 coordinates (Literature 3).	No graph or ladder use.

Evidence record	Matrix use	Graph and ladder use
mostafazadeh-2001 (literature)	9 coordinates (Literature 4, Pieces 5).	No graph or ladder use.
muehlhoff-2010 (literature)	1 coordinates (Local 1).	No graph or ladder use.
nachtergaele-raz-schlein-sims-2007 (literature)	1 coordinates (Pieces 1).	No graph or ladder use.
neumann-pape-streicher-2018 (literature)	14 coordinates (Literature 8, Pieces 6).	No graph or ladder use.
pischke-2025-semigroups (literature)	1 coordinates (Local 1).	No graph or ladder use.
pour-el-richards-1981 (literature)	4 coordinates (Literature 1, Pieces 3).	graph edges 10.
richman-bridges-1999 (literature)	9 coordinates (Literature 4, Pieces 5).	No graph or ladder use.
selivanova-selivanov-2013 (literature)	3 coordinates (Literature 1, Pieces 2).	No graph or ladder use.
selivanova-selivanov-2018 (literature)	2 coordinates (Literature 1, Pieces 1).	No graph or ladder use.
simpson-1984-ode (literature)	1 coordinates (Pieces 1).	No graph or ladder use.
weihrauch-zhong-2002 (literature)	3 coordinates (Literature 1, Pieces 2).	graph edges 10.
weihrauch-zhong-2006-fundamental (literature)	1 coordinates (Pieces 1).	No graph or ladder use.
weihrauch-zhong-2007-cauchy (literature)	1 coordinates (Literature 1).	No graph or ladder use.
zhong-1999-sobolev (literature)	1 coordinates (Literature 1).	No graph or ladder use.
zhong-weihrauch-2003-distributions (literature)	1 coordinates (Pieces 1).	No graph or ladder use.
zohar-burrello-2014 (literature)	22 coordinates (Local 3, Literature 6, Pieces 13).	No graph or ladder use.

The complete normalized dataset is `foundations/site/data.json` (SHA-256 `5e1594303360 8c792f6d979048401f12df77b1b081c8bb8b744d9e1bb8fa00e2`; canonical digest `378a9806111aec 5b00bb9b9d71e8b9bbeaad573feb163684e967aa8cc34de625`). These tables are generated from that file and contain no hand-copied coverage counts, relation edges, evidence roles, boundaries, or citations.

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